

AIRMAN

SERVICE MANUAL

PACKAGED SCREW COMPRESSOR

SAS22UD-58/68

SAS22U-58/68

SAS37UD-58/68

SAS37U-58/68

(50Hz/60Hz)



HOKUETSU INDUSTRIES CO., LTD.

Preface

This service manual explains about the cautions for maintenance jobs and is to serve a guide for the electric system, and troubleshooting for service personnel.

In this book the fundamental matters and other things already mentioned in the “Instruction Manual” and the “Parts Catalogue” are omitted to avoid duplication.

Therefore, for the operation and handling of this unit, we request you to refer to the instruction manual and caution plates, and further for the structure and components of the unit, please refer to the “Parts Catalogue” separately to be supplied with the unit.

If you should find any description which does not coincide with the instruction manual and parts catalog, we request you to make sure to start the job after clarifying it.

Service personnel is required to safely take quick and proper countermeasures as well as to use correct technology of maintenance in case of field services and periodical maintenance.

It is important that service personnel should have proper and sufficient knowledge about the structure and function of the unit and should be well familiar with such technique mentioned in them.

Also this book contains “Safety”. Make sure to observe “SAFETY principles” because if they should be neglected and ignored, it could lead to death accidents as well as serious injury.

When common persons are engaged in operation and servicing works, safety instruction and maintenance services training should be given to them beforehand.

They should read “SAFETY Cautions” mentioned in this book and understand them well so that they should work in safe environment.

Table of Contents

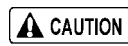
1. Safety	1-1
1.1 Caution before Operation	1-2
1.2 Caution during Operation	1-3
1.3 Caution during Inspection and Maintenance	1-5
1.4 Safety Warning Labels	1-6
2. Specifications	2-1
2.1 Unit Appearance and Part Names	2-1
2.2 Internal Components	2-2
2.3 Compressor Specifications	2-3
2.4 Motor Specifications	2-4
2.5 Size and Dimensions	2-5
3. Cautions on Maintenance Job	3-1
3.1 Cautions	3-1
3.2 Tightening	3-3
4. Periodic Inspection/Maintenance	4-1
4.1 Periodic Inspection / Maintenance List	4-1
4.2 Inspection / Maintenance	4-4
5. Electrical System	5-1
5.1 Controller	5-1
5.2 Electrical Parts	5-9
5.3 Electrical Wiring Diagram	5-15
5.4 Adjustment	5-23
6 Troubleshooting	6-1
6.1 How to Proceed Diagnosis	6-1
6.2 Failure of Capacity Control System	6-4
6.3 Function of Emergency Switch	6-6
6.4 Others	6-12
7 Reference Data	7-1
7.1 Table of Reference Data	7-1
7.2 Piping Diagram	7-2

※ For the details of operation and installation, see Instruction Manual.

1. Safety

This manual explains and illustrates general requirements for safety and cautions for safety.

Read these safety requirements carefully and fully understand the contents before starting the machine.

For your better recognition, according to the degree of potential danger, safety messages are classified into three hierarchical categories, namely, , , and  with a caution symbol —attached to each message.

When one of these messages is found, please take preventive measures for safety to carry out “SAFETY OPERATION AND PROPER MAINTENANCE OF THE UNIT.”



DANGER indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury. This signal word is to be limited to the most extreme situations.



WARNING indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.



CAUTION indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.



IMPORTANT indicates important caution messages for the performance or durability of the unit.

Follow warnings mentioned in this manual. This manual does not describe all safety items. We, therefore, advise you to pay special attention to all items (even though they may not be described in the manual) for your safety.

1. Safety

1.1 Caution before Operation

DANGER

Air pressurized construction method prohibited

- Never use the unit directly or indirectly for the following purposes:
- Never use the unit for respirator equipment by which compressed air is supplied for human consumption. The compressed air contains carbon monoxide and other contaminants, and such air may cause serious injury or even death if used by a person for respiration.
- This compressor is not designed for air pressurized construction method and underwater diving jobs. Never use compressed air for human consumption such as pressurizing diving air tanks. Consumption of compressed air can cause death while diving.

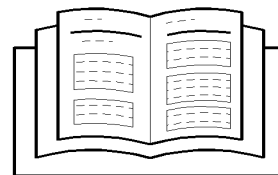


PC005

WARNING

Follow the safety instructions

- Read each instruction plate which is displayed in the manual or on the unit carefully, understand its content and follow the indications thereof.
- Keep the Safety Warning labels clean. When they are damaged or missing, apply new ones.
- Do not modify the machine without prior approval. The safety may be compromised, functions may be deteriorated, or machine life may be shortened.
- Never use the unit for the purpose of compression of gases other than air, or as a vacuum pump. Otherwise, serious accidents may occur.



TR0086

WARNING

Safety outfit

- When handling machine, do not wear;
- Loose clothes
- Clothes with unbuttoned sleeves
- Hanging tie scarf
- Accessories such as dangling jewelry
- Such outfit could be caught in the machine or dragged in the rotating portion of the machine, and this could cause a serious injury.



TR0084

1. Safety



CAUTION

Safety equipments

- Wear helmet, safety glasses, earplugs, safety shoes, safety gloves and a mask, according to the requirements of each operation.



TR0085



CAUTION

Safety fittings

- Have first-aid boxes and fire-extinguishers near the unit ready for emergency situations such as injuries and a fire.
- It is advisable to have a list of phone numbers of doctors, ambulance and the fire department available in case of emergency.



TR0096

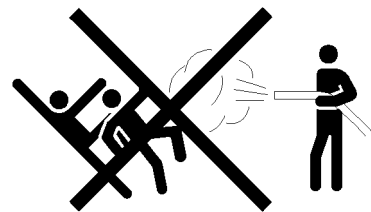
1.2 Caution during Operation



WARNING

Never direct the compressed air to people and foods

- Never blow compressed air directly at people. Scattered impurities, dust, or foreign objects in the compressed air may cause skin and eyes to be seriously injured.
- Blowing compressed air on food is prohibited.



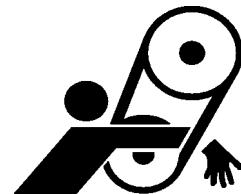
TR0092



WARNING

Hands off from rotating parts and belts

- Keep hands off from the rotating portion or belts while running. It could cause serious injuries if hands should be caught in.



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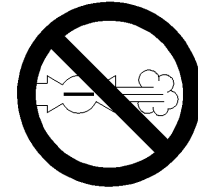
1. Safety



CAUTION

Operation with compressed air supply port opened is prohibited

- Do not operate the machine with service valves and relief valve open unless air hoses and/or pipes are connected. High-pressurized air blows out and its air pressure could cause injury to the people nearby.
- When the machine has to be unavoidably temporarily operated with its port open, be sure to mount a silencer to reduce noise and wear protective materials such as earplugs to prevent damage to hearing.



D003



CAUTION

Do not touch hot parts

- Do not touch hot portions of the machine while inspecting the machine when running. Such parts as pipe and discharging pipe are especially hot, so never touch those parts, because it could cause serious burns.



W005

1. Safety

1.3 Caution during Inspection and Maintenance



WARNING

Hang a "Now Checking and under Maintenance" tag

- Turn the power OFF before starting inspection, and hang a "Now Checking and under Maintenance" tag where it can be easily seen. The checker must keep the key during checking and maintenance.



SY001



WARNING

Refilling and Draining of compressor oil

- When you refill and drain the separator receiver tank with compressor oil, stop the machine, and make sure that the pressure gauge indicates 0MPa and there is no residual pressure in it, and then gradually loosen the oil filler cap for refilling oil.
- In case there is residual pressure in the receiver tank, both extremely hot compressed air and oil could jet out and you may be scalded or seriously injured.



W011



WARNING

Fear of electric shock

- Be sure to keep a wet hand away from the power source or any other electric equipment. Electric shock can cause serious accident or even death.



W013



WARNING

Cleaning by air-blow

- When cleaning dust accumulated in such devices as the air-filter, by blowing compressed air, wear safety glasses, etc. to protect your eyes.



M003



CAUTION

Fear of fire

- Be sure to perform the periodical check of electric equipment such as motor, and compressor oil and oil separator.
- Neglecting checks could cause overheat of the oil, resulting in a fire.



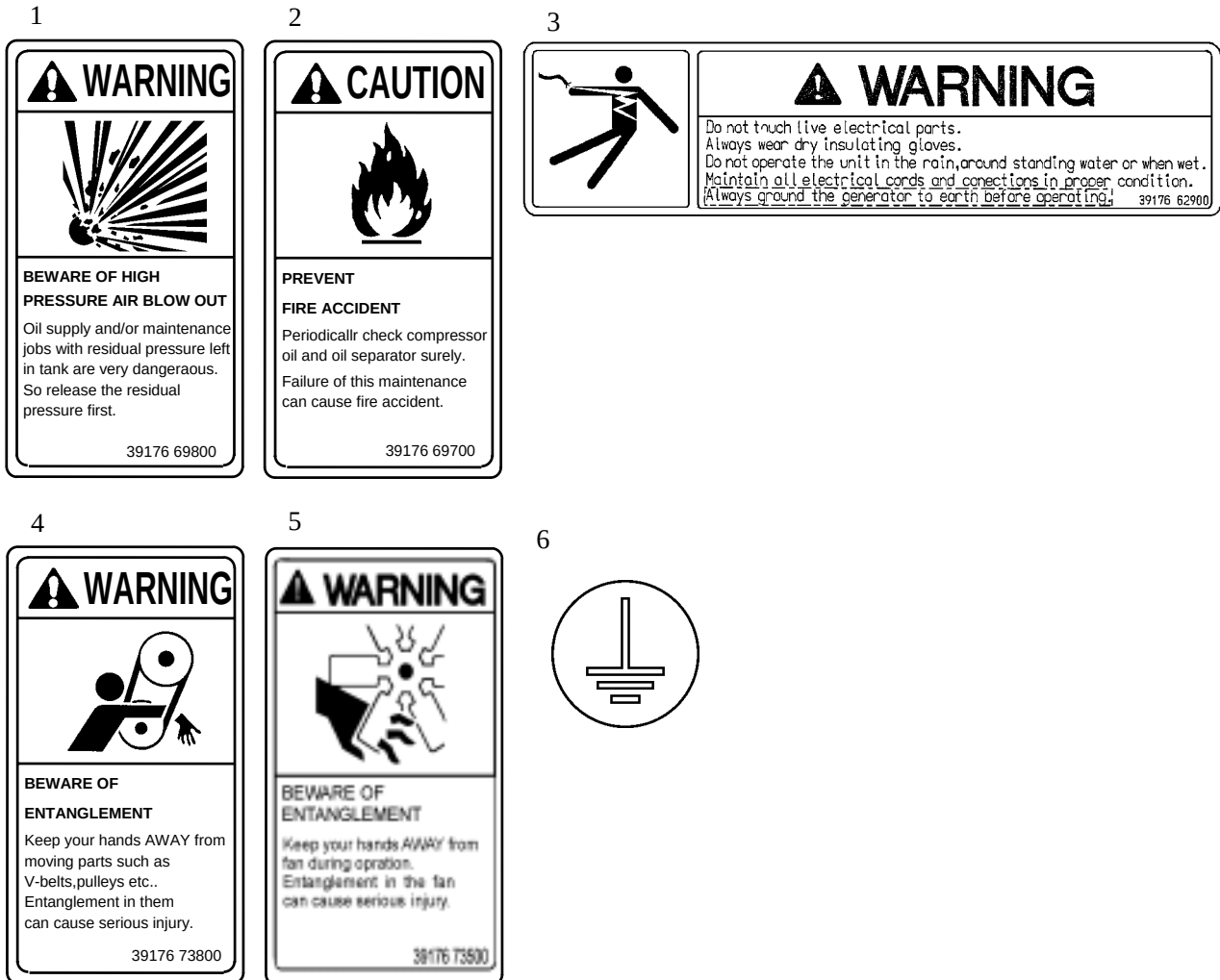
W004

1. Safety

1.4 Safety Warning Labels

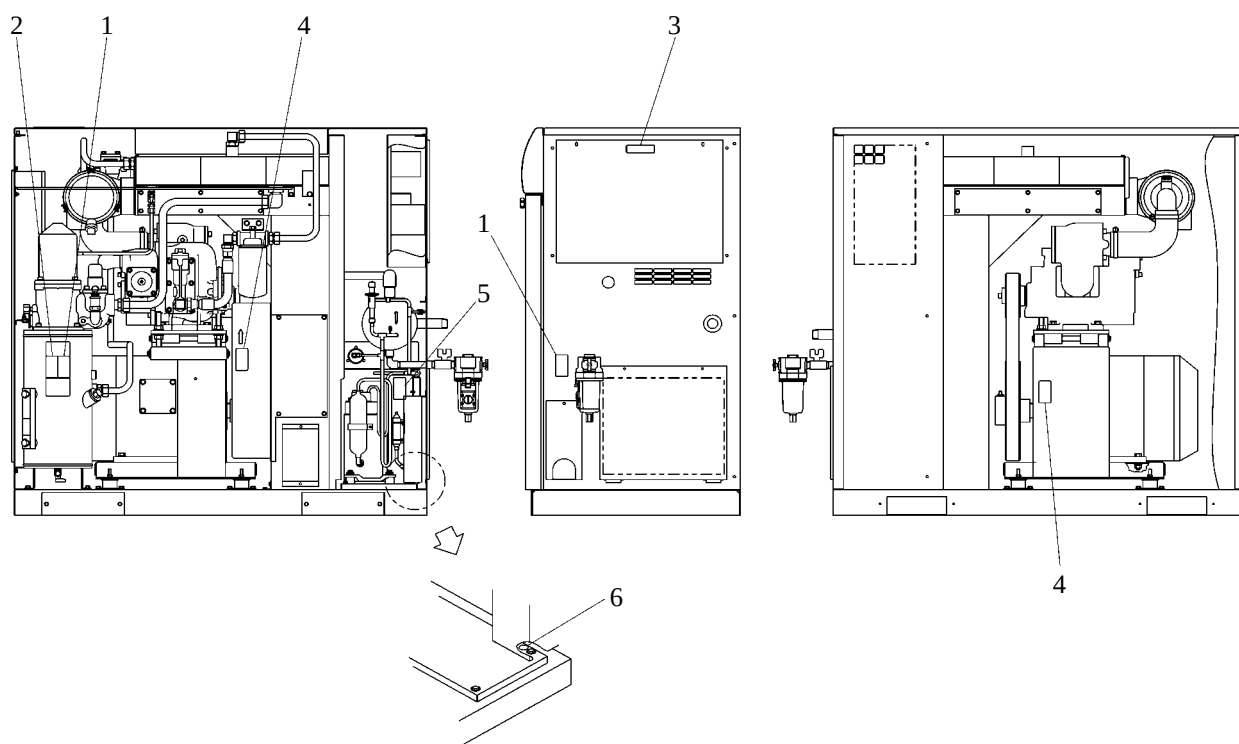
Following labels are attached to the machine.

Keep them clean all the time. If they are damaged or peeled off, immediately place an order with your nearest dealer for replacement, with the number indicated on the lower right corner of the label. Adhere a new one to the original place.



1. Safety

● Locations of warning labels attached.

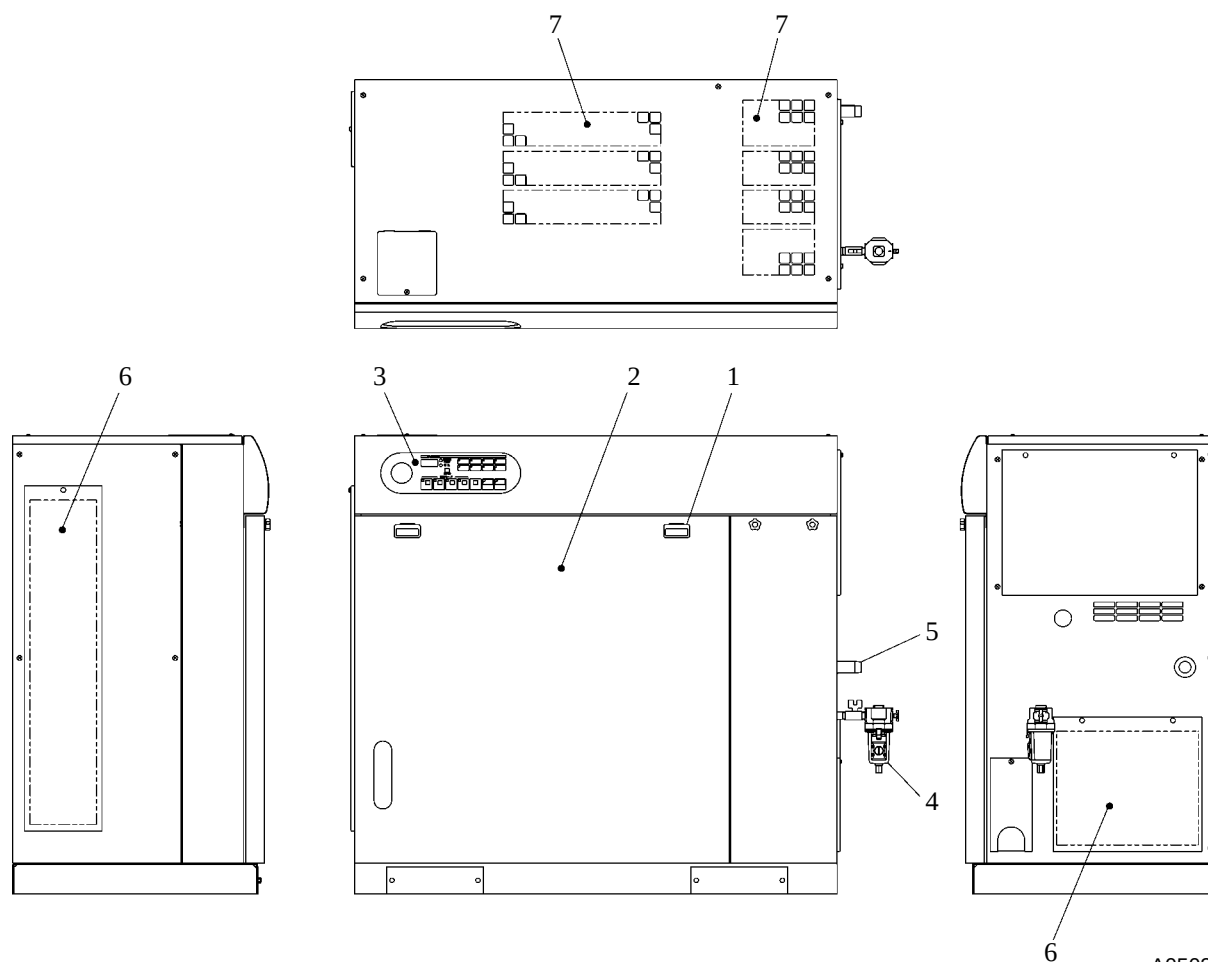


SAS22UD-58/68 (Dryer built-in Type)

A050471E

2. Specifications

2.1 Unit Appearance and Part Names

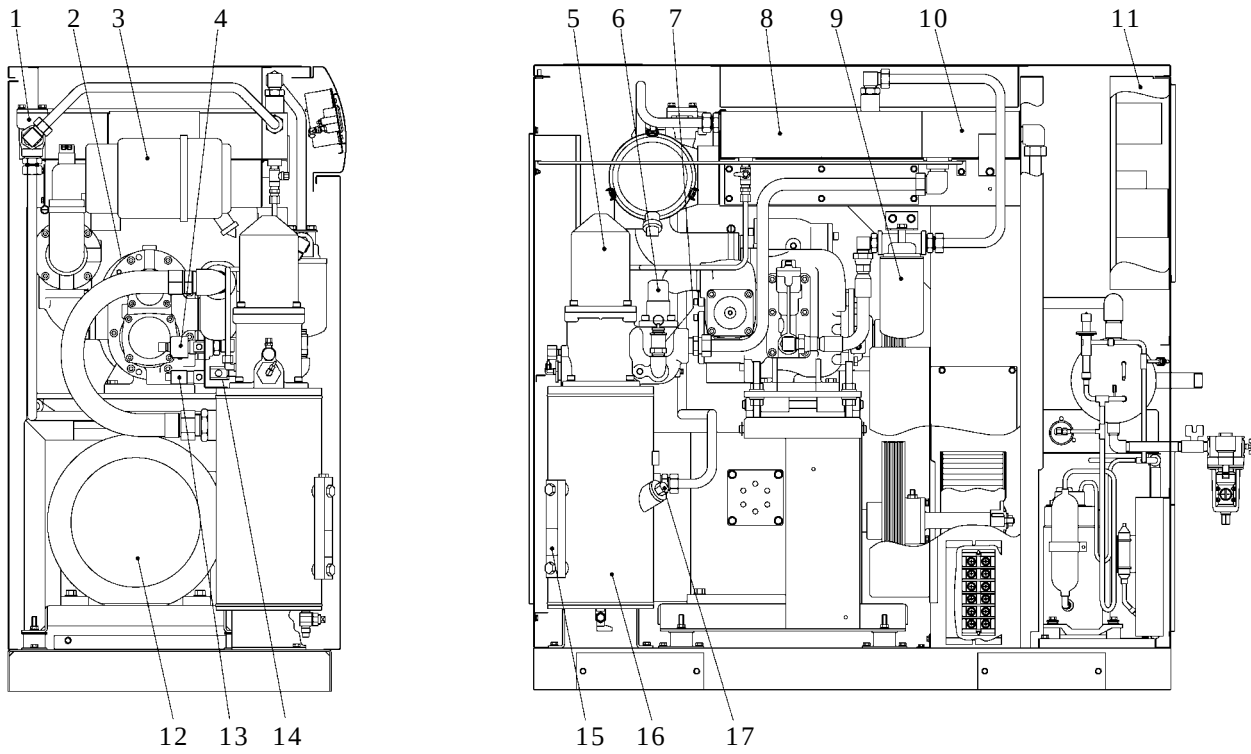


SAS22UD-58/68 (Dryer built-in Type)

- | | |
|---------------------|----------------------------|
| 1. Handle | 5. Discharge port 25A (1B) |
| 2. Door | 6. Cooling air inlet |
| 3. Operation panel | 7. Cooling air outlet |
| 4. Auto-drain valve | |

2. Specifications

2.2 Internal Components



A050466

SAS22UD-58/68 (Dryer built-in Type)

- | | |
|--|---|
| 1. By-pass valve | 10. After cooler |
| 2. Air-end | 11. Starter panel |
| 3. Air filter | 12. Motor |
| 4. Solenoid valve for capacity control | 13. Solenoid valve for separator recovery |
| 5. Oil separator | 14. Solenoid valve for auto-relief |
| 6. Pressure control valve | 15. Oil level gauge |
| 7. Safety valve | 16. Separator receiver tank |
| 8. Oil cooler | 17. Oil filler port |
| 9. Oil filter | |

2. Specifications

2.3 Compressor Specifications

2.3.1 Dryer built-in type

Compressor	Model		SAS22UD-58		SAS22UD-68		SAS37UD-58		SAS37UD-68	
	Type		Rotary-twin screw, single stage, oil cooled							
	Free air delivery	m³/min	3.7	2.9	3.7	2.9	6.1	5.2	6.1	5.2
	Working pressure	MPa	0.69	0.83	0.69	0.83	0.69	0.83	0.69	0.83
	Capacity control system		AIRMAN Unloader (Suction port closing, Discharge chamber pressure relief, Oil volume control)							
	Lubrication system		Forced oil injection by compressed air pressure							
	Driving system		Belt drive							
	Lubrication oil		All Seasons							
	Lubrication oil capacity	L	13.5				24			
	Tank capacity	m³	0.017				0.028			
Motor	Type		Three-phase squirrel cage induction/totally-enclosed fan cooled							
	Power output	kW	22				37			
	Starting system		Star-Delta (3-Contactors)							
Air outlet port			25A(1B)				40A(1 1/2B)			
Dryer	Dew-point of discharged air	℃	Under the ambient pressure less than 10							
	Power consumption	kW	1.15		1.43/1.43		1.5		1.9/1.9	
	Refrigerant		R407C (Refrigerant enclosed: 360g)				R407C (Refrigerant enclosed: 800g)			
Approx. dimensions	Width	mm	1,380				1,620			
	Length	mm	720				880			
	Height	mm	1,350				1,460			
Approx. dry weight		kg	660				940			
Approx. weight for maintenance		kg	670				960			

- Free air delivery is a value converted into a suction status at air temperature of 30°C.
- Dew point of discharge air is the value at the ambient temperature of 30°C.
- When the Dryer discharges sediment, the air quantity will decrease due to the deposit of sediment.

2. Specifications

2.3.2 Without dryer type

Compressor	Model		SAS22U-58		SAS22U-68		SAS37U-58		SAS37U-68	
	Type		Rotary-twin screw, single stage, oil cooled							
	Free air delivery	m³/min	3.7	2.9	3.7	2.9	6.1	5.2	6.1	5.2
	Working pressure	MPa	0.69	0.83	0.69	0.83	0.69	0.83	0.69	0.83
	Capacity control system		AIRMAN Unloader (Suction port closing, Discharge chamber pressure relief, Oil volume control)							
	Lubrication system		Forced oil injection by compressed air pressure							
	Driving system		Belt drive							
	Lubrication oil		All Seasons							
	Lubrication oil capacity	L	13.5				24			
	Tank capacity	m³	0.017				0.028			
Motor	Type		Three-phase squirrel cage induction/totally-enclosed fan cooled							
	Power output	kW	22				37			
	Starting system		Star-Delta (3-Contactors)							
Air outlet port			25A(1B)				40A(1 1/2B)			
Approx. dimensions	Width	mm	1,380				1,620			
	Length	mm	720				880			
	Height	mm	1,350				1,460			
Approx. dry weight		kg	620				880			
Approx. weight for maintenance		kg	630				900			

● Free air delivery is a value converted into a suction status at air temperature of 30°C.

2.4 Motor Specifications (Totally-enclosed fan cooled type)

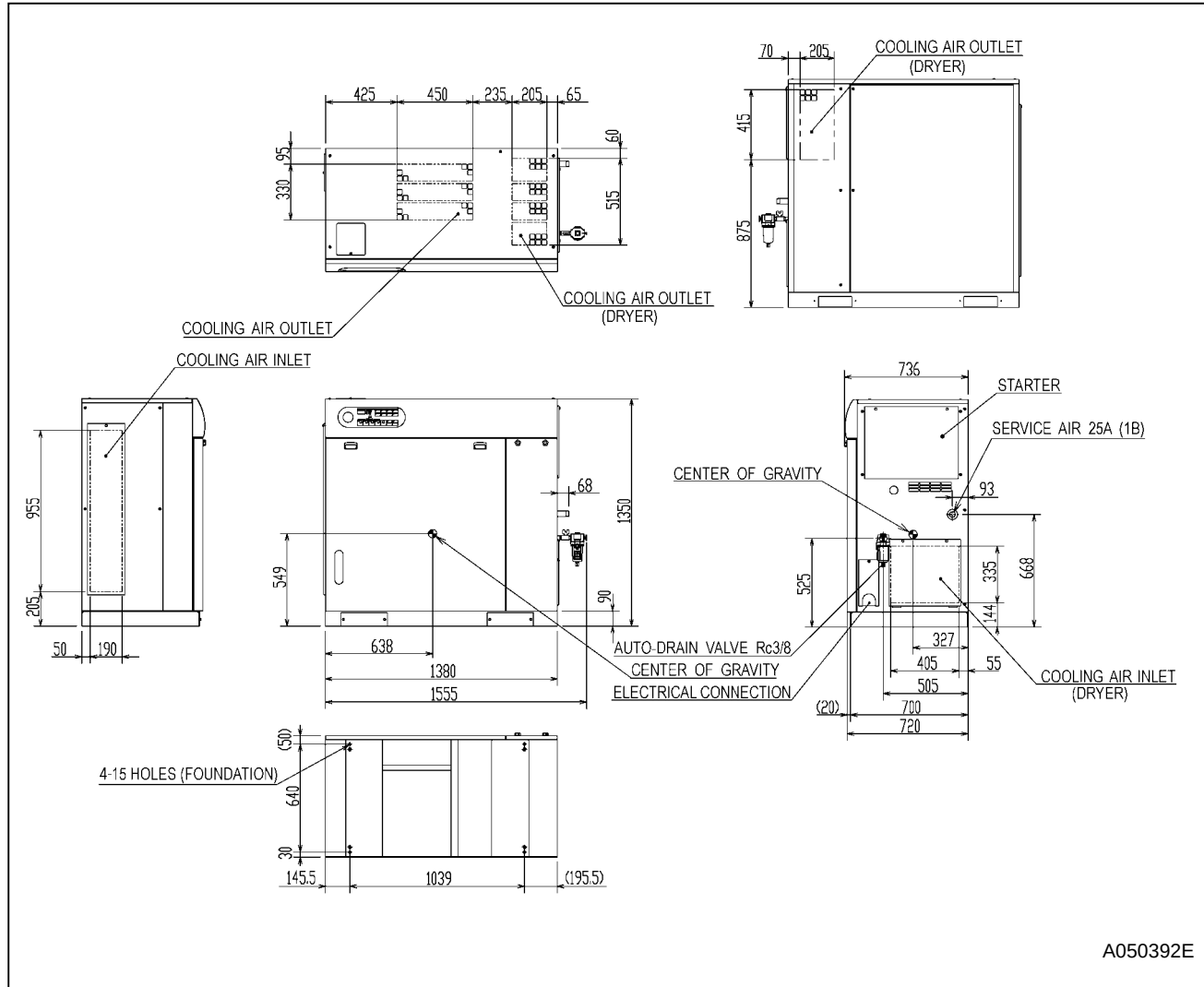
Model		SAS22U(D)-58	SAS22U(D)-68	SAS37U(D)-58	SAS37U(D)-68
Motor Type		TOSHIBA TIKK-FBK21A			
Power output kW		22		37	
200V	Frequency Hz	50	60	50	60
	Power source V	200	200/220	200	200/220
	Current A	93	88/83	156	147/138
	Number of poles	4P			
	Insulation	F			
400V	Frequency Hz	50	60	50	60
	Power source V	400	400/440	400	400/440
	Current A	46.5	44/41.5	78	73.5/69
	Number of poles	4P			
	Insulation	F			

2. Specifications

2.5 Size and Dimensions

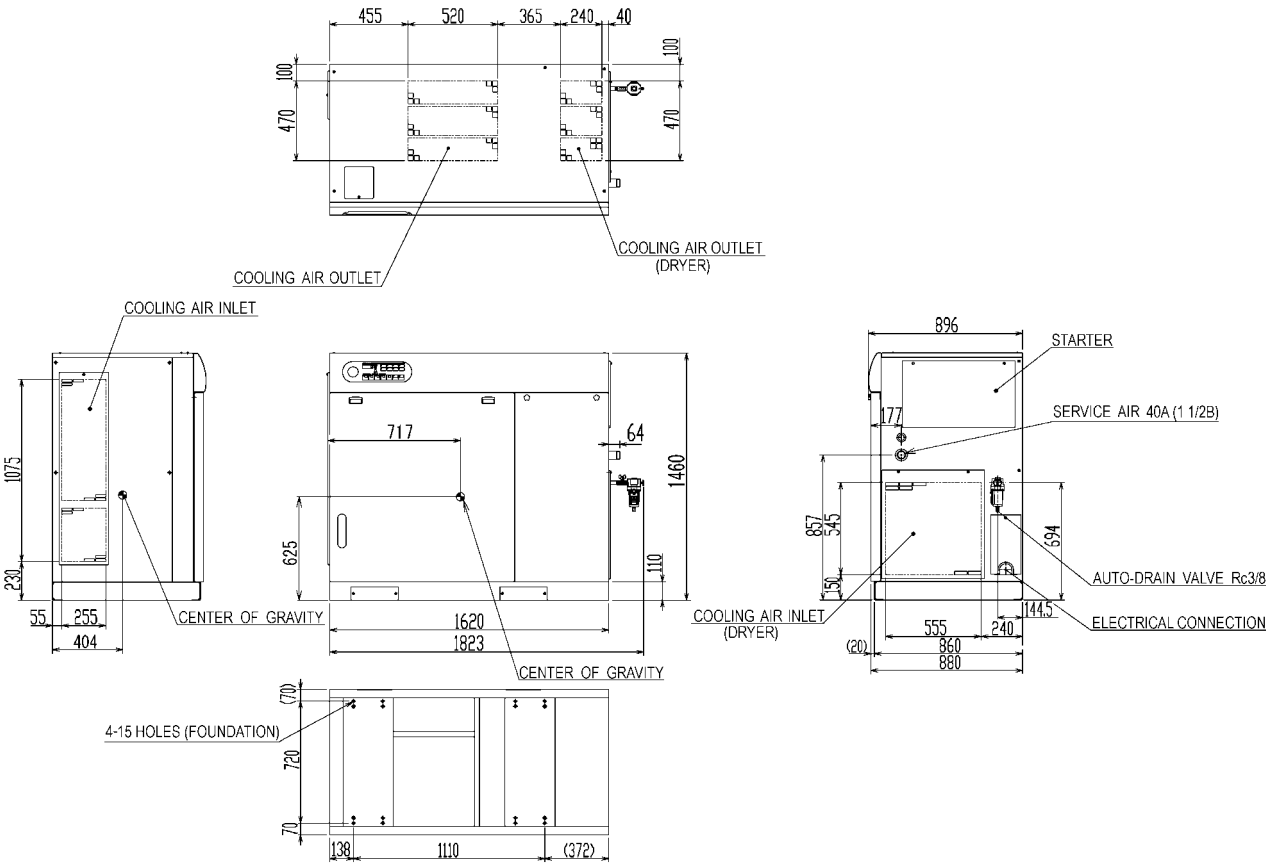
2.5.1 Dryer built-in type

SAS22UD-58/68



2. Specifications

SAS37UD-58/68

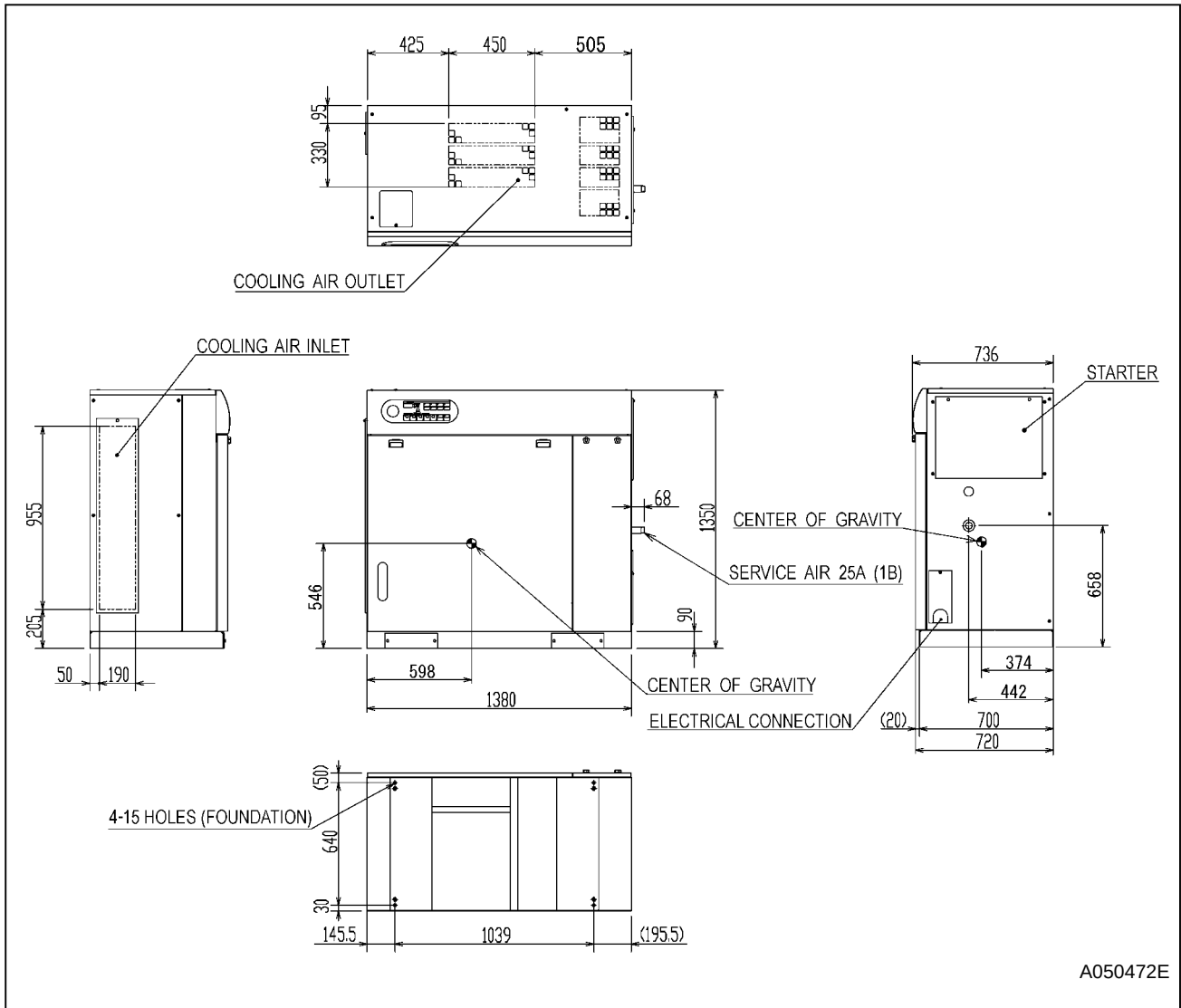


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2. Specifications

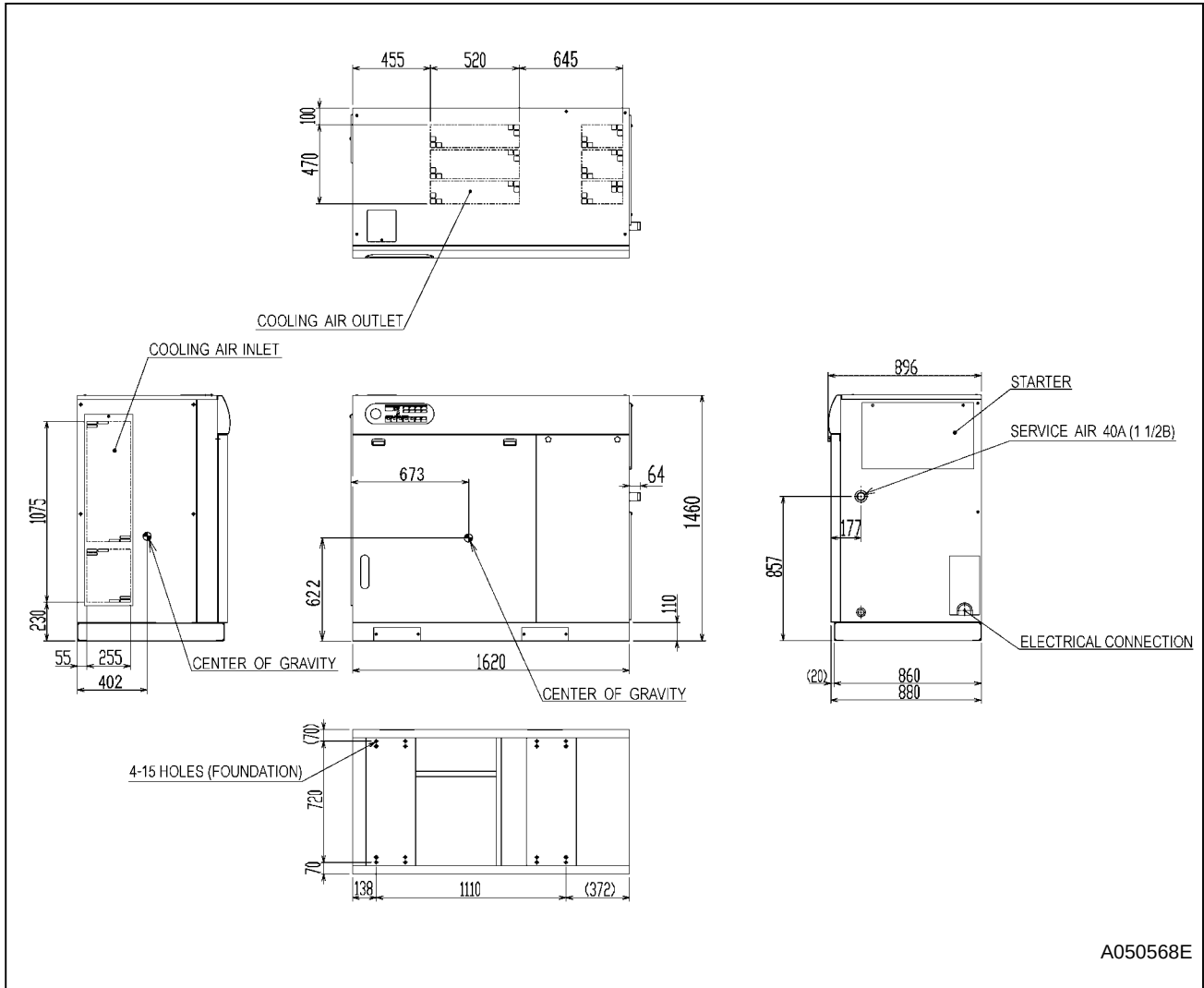
2.5.2 Without dryer type

SAS22U-58/68



2. Specifications

SAS37U-58/68



3. Cautions on Maintenance Job

3.1 Cautions

3.1.1 Preparations before maintenance job

(1) How to perform maintenance job

It is important to perform maintenance jobs, always considering what kinds of equipments, tools, materials and oils should be used, what portions or parts of the machine should be inspected or adjusted or disassembled, and also what is the important points of the maintenance jobs.

(2) Prevention of oil overflow

In order to prevent oil from overflowing in case of compressor oil change and removal or installation of oil piping, make sure to prepare an oil reservoir. And also if an excessive oil flow is expected beforehand, it is necessary to empty the receiver tank beforehand.

(3) On removal or installation of parts or assemblies

In case of removing complicated parts or components, mark the match mark for reinstallation.

(4) Equipments to be prepared beforehand

① Measuring instruments

Thickness gauge, Torque wrench (Preset type or spanner type)

② General tools

③ Jigs and special tools

④ Literature (Instruction manual, parts catalog and manual etc...)

⑤ Replacement parts

⑥ Auxiliary materials

Item	Name	Application
Cleaning agent	Cosmo Flush C	Compressor Air-end
Grease	Epnoc AP-1	O-ring
Sealant for screwing-in	Loctite No.577	Screwing-in, block plug
Liquid packing for cast iron	Loctite No.515	Inlet, power side, discharge casing
Liquid packing for aluminum (hardening time :48Hr)	Loctite FMD-127	Unloader
Liquid packing for aluminum acceleration hardening (hardening time :2Hr)	Loctite primer N	Unloader
Compressor oil	All Seasons	Inside the cylinder
Diesel oil for cleaning		Cleaning
Thinner		Cleaning
Release agent for liquid packing	Loctite Gasket remover	Portion at which liquid packing is coated
Oil stone		Portion at which liquid packing is coated
Rag		

⑦ Equipments

Hot oil bath

Chain block

Impact wrench

3. Cautions on Maintenance Job

3.1.2 Cautions during disassembly and assembly

- ① Have to choose place where the machine is disassembled for maintenance job shall be clean, good ventilation and have much space to do it.
- ② Have to clean and remove dust from disassemble equipment.
- ③ Pay attention ventilation and disaster prevention when cleaning disassembled parts by diesel oil.
- ④ Before removing nylon tubes or hoses, have to clean dust and after removing them, make necessary treatment to prevent dust or foreign matters from entering inside.
- ⑤ Make sure to check the disassembled parts for any dirt and decoration, and then wash them by diesel oil. But, don't soak rubber sort to diesel oil absolutely.
- ⑥ After washing, put it on the clean paper or cloth to arrange in disassembled order and then do not damage them or do not make dirty.
- ⑦ As the parts or components are precision-made, do not damage them by falling or colliding.
- ⑧ Don't use disassemble O-ring, filter, oil. (have to abrogate it). Have to use new parts for re-assembling.
- ⑨ When reassembling, place the parts according to the assembling turn and check to prepare all parts. (without any part left unused.)
- ⑩ If there are any contact parts used, use parts assemblies.
- ⑪ Install O-ring with new lithium based grease coated on.
- ⑫ Make sure to check tightening torque and clearance when assembling the disassembled parts.
- ⑬ When disassembling and reassembling the machine, have to use Jigs and tools correctly. If you use it unfit, it is cause to break machine and tools or to get injured.
- ⑭ When stop disassembling halfway, it is cause to occur rust or soil them by humidity or rubbish. If it is necessary to stop disassembling job on the way, make sure to protect them against rust and dirt.
- ⑮ After finishing assembling job, make sure to check following.
 - Oil leakage
 - To check tightening torque of every bolt
 - The direction of rotation
 - Part left behind to assembling
 - Check oil level of receiver tank
- ⑯ After finishing re-assembling job, make sure to carefully perform trial operation so that it may not cause seizure or overheating to the unit.
- ⑰ When will perform disassembling or assembling by several person, have to obey suggestion of leader. And when go to next progress, to check safety to call out to each other.

3. Cautions on Meintenance Job

3.2 Tightening

3.2.1 Tightening torque

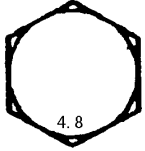
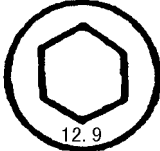
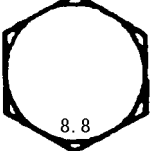
Tightening place				Bolt diameter (mm)	Width of across flat (mm)	Tightening torque
						N・m (kgf・cm)
Compressor Air-end	Discharge casing and check valve socket bolt			10	8(Hexagon haul)	49 (485)
	Shaft seal cover socket bolt			8	6(Hexagon haul)	25 (245)
	※ End cover and drain pump socket bolt			8	6(Hexagon haul)	12 (124)
	※ Bearing pushing plate socket bolt	Male rotor side		6	5(Hexagon haul)	10 (100)
		Female rotor side	SAS22UD	10	8(Hexagon haul)	49 (485)
			SAS37UD	8	6(Hexagon haul)	25 (245)
	Cylinder cover (oil filler plate)			8	6(Hexagon haul)	25 (245)
	Unloader socket bolt			10	8(Hexagon haul)	49 (485)
Compressor Air-end fixing bolt, nut				12	19	43 (425)
Motor fixing bolt, nut		SAS22UD		12	19	43 (425)
		SAS37UD		16	24	106 (1055)
Separator cover socket bolt				10	8(Hexagon haul)	49 (485)
Separator casing socket bolt				10	8(Hexagon haul)	49 (485)
V-pulley pushing socket bolt (Compressor Air-end side)				10	8(Hexagon haul)	49 (485)
Level gauge bolt				14	22	26 (255)
Sirocco fan shaft idler bolt				8	13	25 (245)
Sirocco fan lock nut				12	19	43 (425)

※ : Make sure to tighten the socket bolts fixing end-cover and bearing retainer plate, just according to the tightening torque specified above without fail. Excessive tightening or insufficient tightening sometimes could cause earlier damage to the bearings and it could loosen the bolts. Take special care for tightening them.

3. Cautions on Maintenance Job

3.2.2 General bolts and nuts tightening torque

When reassembling or rebuilding, tighten bolts and nuts, according to the following torque values in the table.

Kind Strength and sorting Width of across flat Tightening torque Bolt diameter (mm)	Low or Middle carbon steel bolt (SS400B etc...)		High tensile strength bolt (SCM435 etc...)		
	4.6~6.8 (4T~6T)		8.8~12.9 (7T~12T)		
	 Hexagon bolts		 Socket bolts	 Hexagon bolts	
	Hexagon bolts Width of across flat (mm)	Tightening torque N·m (kgf·cm)	Socket bolts Width of across flat (mm)	Hexagon bolts Width of across flat (mm)	Tightening torque N·m (kgf·cm)
6	10	5 (51)	5	10	10 (100)
8	13	12 (124)	6	13	25 (245)
10	17	25 (245)	8	17	49 (485)
12	19	43 (425)	10	19	85 (845)
14	22	68 (675)	12	22	135 (1350)
16	24	106 (1055)	14	24	210 (2100)
18	27	145 (1450)	14	27	290 (2900)
20	30	205 (2050)	17	30	410 (4100)
22	32	280 (2800)	17	32	560 (5600)
24	36	345 (3450)	19	36	710 (7100)
Using place	Bonnet flame etc...		Compressor Air-end, receiver tank and other designation part		

IMPORTANT

- The above torque values in the table shall be applicable for the bolts and nuts used for compressor.
- In general, the above torque values are applicable for the bolts and nuts used for compressor, but such parts used for special portions need be tightened according to the specified torque values indicated.
- When tightening parts or components, make sure to remove rust and dust.

3.2.3 Terminal tightening torque

Should be tighten terminal of electric contactor, thermal relay other electric equipment etc. according to following torque.

Screw size	Tighten torque
	N·m (kgf·cm)
M3.5	1.0 (10)
M4	1.5 (15)
M5	3 (30)
M6	5 (51)
M8	10 (100)
M10	18 (175)
M12	35 (350)

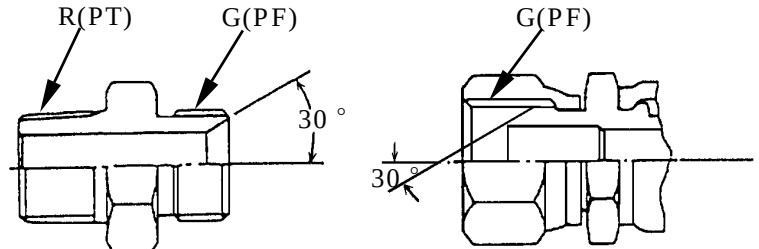
3. Cautions on Maintenance Job

3.2.4 Tightening torque for hose, joint etc...

(1) Union joint

Should be tighten union joint according to following torque.

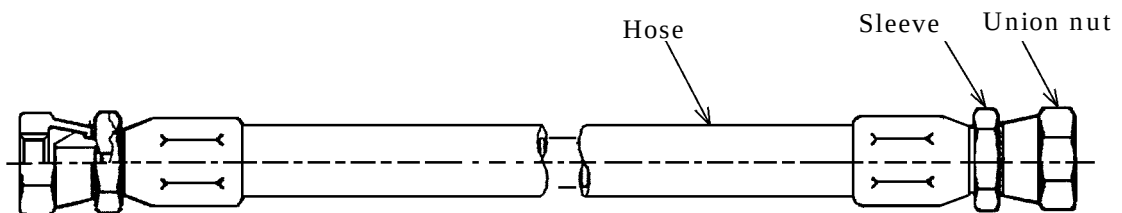
Parallel thread for pipe		
Screw size G (PF)	Tool Width of across flat (mm)	Tightening torque
		N•m (kgf•cm)
1/4	19(17)	25 (250)
3/8	22(19)	50 (500)
1/2	27(22)	59 (600)
3/4	36(30)	120 (1200)
1	41(36)	140 (1400)
1 1/4	50	170 (1700)
1 1/2	60	210 (2100)



TXT104

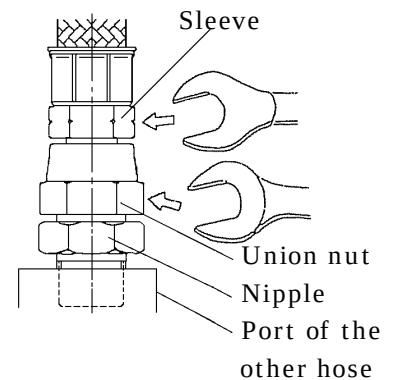
How to install AQP hoses

AQP hoses are used for supplying oil to compressor air end. For the installation of AQP hoses, follow the under-mentioned procedures to prevent hoses from twisted.



Installation procedures

- ① Install union nut to the nipple fitted at the port of the other hose.
Make sure to tighten the union nut with a hand till it reaches "Stop" end of the nipple.
- ② Use two spanners. Hold the sleeve with the one spanner and tighten the union nut with another spanner according to the specified torque.



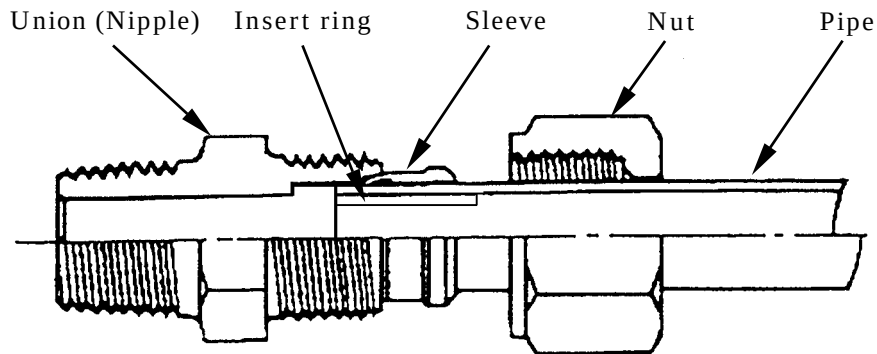
IMPORTANT

- Excessively tightening of union nut can cause excessive wedging action so that union nut can be broken. So tighten the nut, according to the specified torques.
- Damaged seating surface can cause oil leakage. So take special care when disassembling and reassembling.

3. Cautions on Maintenance Job

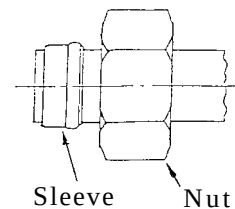
(2) Bite type union

The bite type union used for this pipe is to seal completely high pressure oil. Because the cutting edge of sleeve bites the steel pipe firmly and it is constructed to endure shock pressure and vibration. Therefore, this union is widely used for many hydraulic devices. But bear in mind that oil leak will occur if you are not familiar with the structure and usage of the union. Fit it according to the following steps.

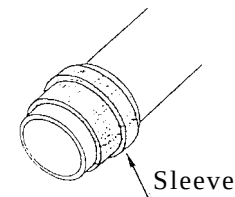


1) Check and confirm the pipe

- ① Check whether this pipe is provided with insert ring, sleeve and nut. (Some insert rings are not used to be inserted, depending on the diameter.)

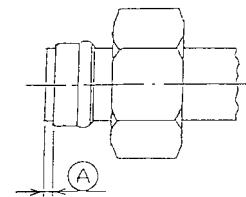


- ② Check the direction of the sleeve and make sure it is not caulked diagonally.



- ③ Visually make sure that there is a clearance of a few millimeters from the end of pipe to the sleeve (size A).

3-4 mm is recommended



IMPORTANT

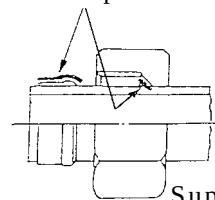
- When retightening the nut after loosening the tightened nut, mark "match-mark" on the location of the nut with white chalk before loosening it and then remove it. When tightening it again, tighten the nut a bit more tight than the position tightened before.

3. Cautions on Maintenance Job

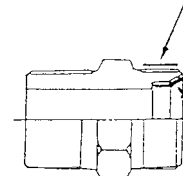
2) Coat pipe and nipple with lubrication oil

- ① Coat the sleeve and nut of pipe with lubricant (NICHIMORI LAP spray or equivalent).
- ② Coat the threaded portion and seat face of nipple with lubricant.

Supply oil to these points



Supply oil to these points



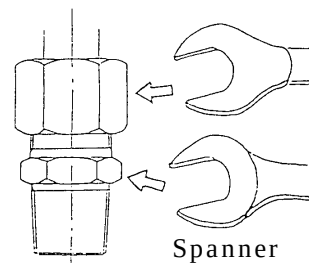
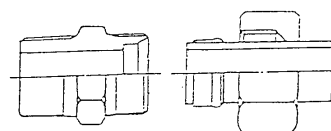
IMPORTANT

- When tightening it without coating lubricant, galling could take place. Therefore, the sleeve turns together. So then it can cause oil leak.

3) Tightening steps

- ① Tighten the pipe to the nipple.
Make sure to tighten the pipe nut with a hand till the nipple "Stop" position.
- ② Tighten the nut of pipe by one spanner with holding the nipple by another spanner.
- ③ Continue to tighten the nut and when you will find sharp torque ruse point, tighten it 1/4 more to make the inside sleeve curves.

Tighten the nut with a hand



Spanner

3. Cautions on Maintenance Job

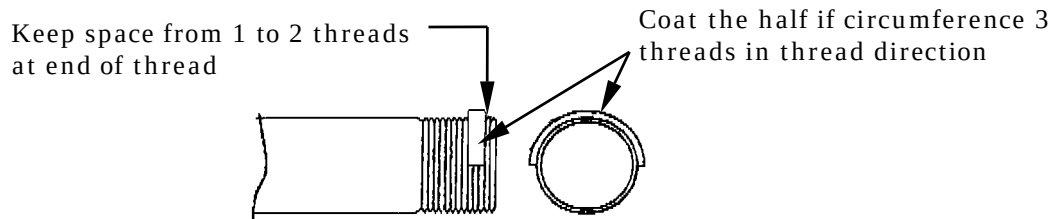
(3) Pipe tapered threads

Sealing tapes are used to prevent any leakage of oil, water, fuel and air by stopping the space between female and male threads of metal pipe and fittings (PT thread for pipe).

Be noted that how to seal varies upon the diameter of thread.

1) How to seal for tube screw less than $\frac{3}{8}$ of screw diameter.

- ① Wipe out oil and fat content with wasted cloth.
- ② Coat sealant "Loctite 577" as shown below.



S_M030

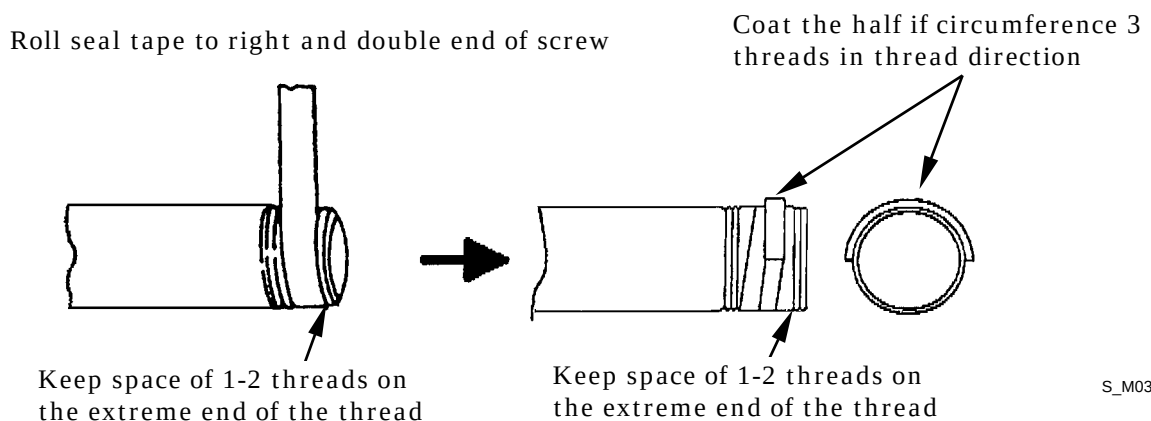
IMPORTANT

- Sealant only can damage sealed portion after the sealant is hardened, and it can cause leakage, if the joint connector is moved. Should the joint be moved, repeat this treatment.

In case of different metals between female thread and male thread (for example, female thread : aluminum male thread : steel etc.) even if the thread diameter is less than $\frac{3}{8}$, apply seal tape (Three Bond TB4501) as well as coat sealant. (Sealant could be cut due to the different coefficient of thermal expansion.)

2) How to seal for tube screw larger than $\frac{1}{2}$ of thread diameter.

- ① Wipe out oil and fat content with wasted cloth.
- ② Roll seal tape (Three Bond TB4501) .
- ③ Coat sealant "Loctite 577" on the seal tape which is attached. According to the figure below.



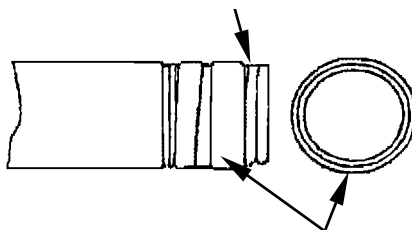
S_M031

3. Cautions on Meintenance Job

3) How to seal for tube screw of larger than 1 1/4 (32A) of thread diameter.

- ① Wipe up oil and fats content from the threaded portion with rag.
- ② Roll seal tape (Three Bond TB4501) . [same as 2] clause]
- ③ C oat sealant “Loctite 577” on seal taped portion as shown below.

Keep space 2 to 3 threads at end of thread



Keep space more than 3 threads coat sealant fully on the circumference

S_M032

(4) Standard tightening torque of piping Pt screw (Reference value)

Unit:N・m(kgf・cm)

Nominal diameter	Tightening torque	Nominal diameter	Tightening torque	Nominal diameter	Tightening torque
1/8	11(110)	3/4	65(660)	2	216(2200)
1/4	22(220)	1	108(1100)	2 1/2	270(2750)
3/8	32(330)	1 1/4	129(1320)	3	323(3300)
1/2	43(440)	1 1/2	162(1650)		

Note : Over torque tightening could lead to deforming of threads and resulting in oil leakage. (2 threads should be left as standard)

3. Cautions on Meintenance Job

(5) Nylon tube

To size of nylon tube are used to this machine. One is Inch size and other one is meter size. Each nylon tubes are not compatible.

Inch size It is use spanner when tightening sleeve nad nut to piping.

Meter size Install to one tach joint.

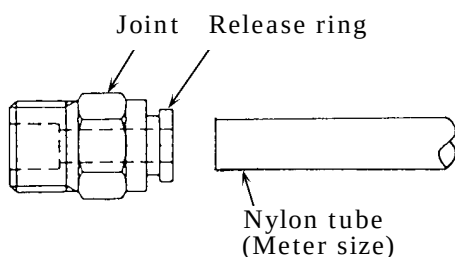
- Some inch sized nylon tubes can be inserted into one tough joint connectors... But sometimes it is difficult to remove them. Be careful not to use inch sized nylon tubes.
- Circumference of each nylon tube serves to prevent leaking. When any damage or deflectionon the circumference is found, cut such damaged portions of the nylon tube.

Each nylon tube is marked nominal size in a constant distance on it, and so it is easy to distinguish such portions... But the circumference of nylon tube is quite same. So be careful not to make a mistake in its installation.

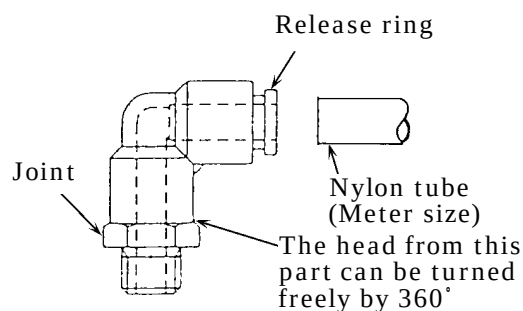
Cautions on how to use one tough joint connector.

- Rubber seal isprovided inside the joint connector. As the joint is repeatedly installed and removed many times, air leak sometimes can take place.
- Installation and/or removal must not be performed under pressurization.
- Installing any nylon tube, you find it difficult to confirm whether it is fully and correctly installed, looking from outside. After it is installed, make sure to check that it is surely installed and cannot be pulled out, by poulling it.
- For removal of nylon tube, press the nylon tube to the fitting side and push the release ring of one touch joint to the fitting side, then the locked condition is released so that it can be extracted easily.

Straight nipple

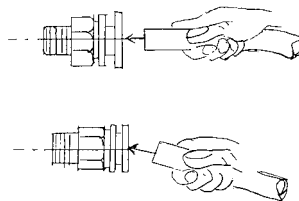


Nipple elbow



IMPORTANT

- As shown in the sketch, put the nylon tube straight against the inlet of the fitting. If the tube is installed diagonally, the internal diameter of lock ring gets wider and it can not hold the tube and it drops.



3. Cautions on Maintenance Job

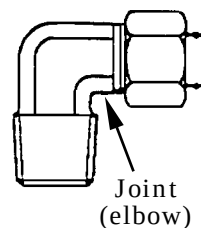
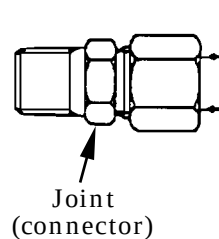
(6) Tightening for joint of nylon tube & sleeve nut

Following chart is for tightening torque of each size joint and sleeve nut.

1) Tightening of Joint

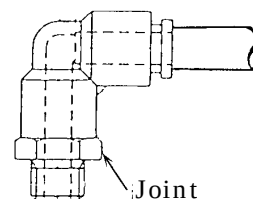
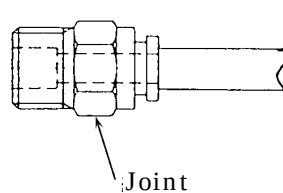
Screw size R (PT)	Tightening torque
	N•m (kgf•cm)
1/8	7.8 (80)
1/4	13 (130)
3/8	23 (230)
1/2	28 (290)

[Inch size]



TXT106

[Meter size]



IMPORTANT

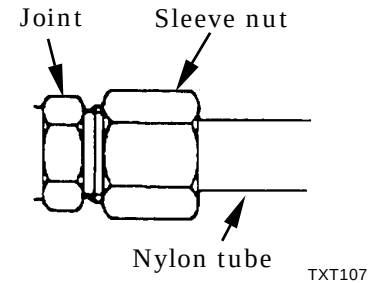
- Tightening torque is specified for each size thread.
- In case white coated portion is peeled of, apply sealing tape to the portion, according to the next clause.

3. Cautions on Maintenance Job

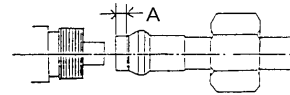
2) Tightening sleeve nut

Installing new nylon tube, sleeve and sleeve nut, attach the nylon tube firmly to the joint connector and tighten the sleeve nut till proper revolutions by hand, and them tighten the sleeve nuts till proper turns, using a tightening tool such as spanner.

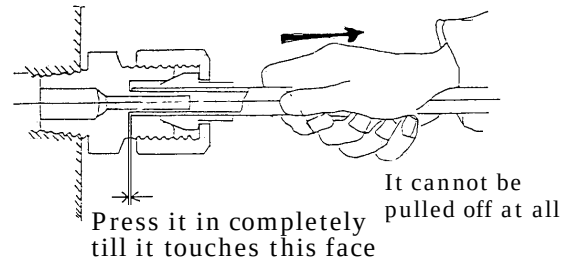
Tube nominal diameter	Tool width across flat (mm)	Sleeve nuts proper tightening turns (turn times)	Tightening torque
			N·m (kgf·cm)
1/8	8	1/2~1	0.8 (9)
3/16	10	1/2~1	1.8 (18)
1/4	12	1/2~1	4 (41)
3/8	17	1/2~1	8.3 (85)
1/2	19	1/2~1	12 (120)



Before installing nylon tube, check the size of A. If you do not use caulking tool, check the size of A again after removing it once.



After having tightened it, make sure to confirm that the tube can not be removed by pulling it off.



IMPORTANT

- In order to tighten sleeve nut correctly, keep the correct tightening turns of sleeve nut in accordance with the nut turning constant times.
The optimum turning times mean the revolutions times after tightening the nuts fully by hand.
The tightening torque of above chart is referent value.

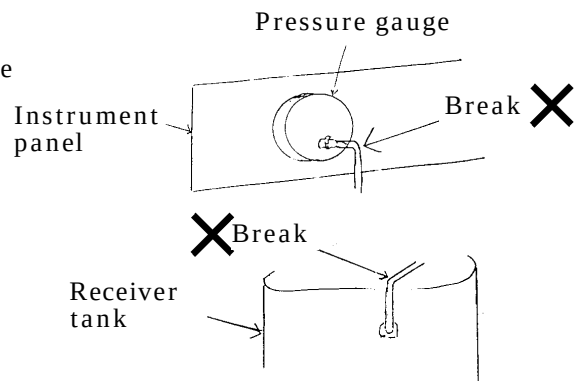
3. Cautions on Meintenance Job

(7) How to install hose

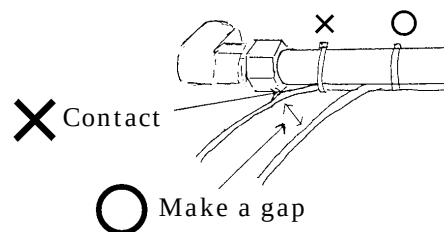
Pay attation followig part when install hose and nylon tube.

1) Nylon tube

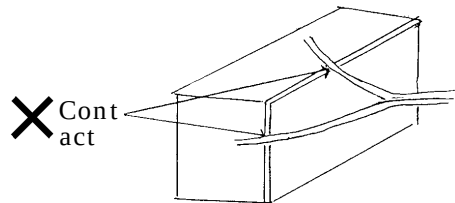
- ① Check 3/16 size nylon tube for any break. (Pipe for pressure gauge)



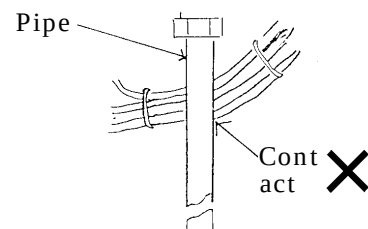
- ② Connect nylon tube, preventing it from contacting any edge of copper pipe and steel pipe nut.



- ③ Make connection of nylon tube, taking care not to touch the edge of bracket and plate etc.



- ④ Make connection of nylon tube and flexible pipe, taking care not to touch any copper or steel pipe. (If it is avoidably impossible, use it, being protected by felt)



S_M121

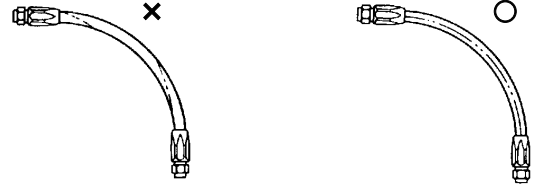
3. Cautions on Maintenance Job

2) Hose and flexible pipe

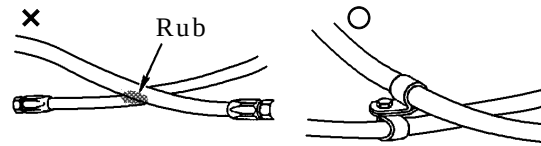
- ① Authentic hose and flexible pipe when replacing them. If not, it will cause oil leakage, break and disconnection, and it leads to fire.

- ② Never use hose and flexible pipe twisted.

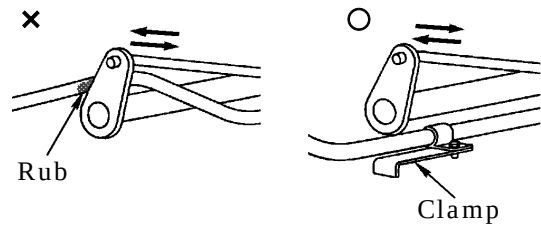
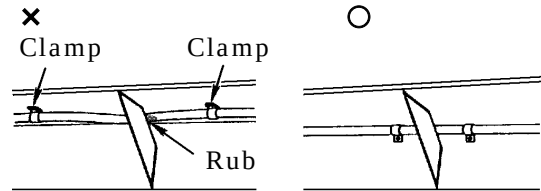
Twisted hose or flexible pipe will cause oil leakage, break and fitting disconnection when they are loaded with high pressure, vibration and shock. The printed mark on outside surface of hose is a guide for prevention of twisted connection.



- ③ Avoid contact between hose and flexible pipe. If the hose and flexible pipe are rubbing each other continuously, it causes damages and also burst on hose and flexible pipe.



- ④ Take care not to make hose and flexible pipe touch sharp edge or moving part of the unit or component.



4 Periodic Inspection/Maintenance

4.1 Periodic Inspection/Maintenance List

Regarding the items or parts marked ○ in the table of periodical maintenance customers or users should perform periodical maintenance of the items or parts marked ○ by themselves. For such items or parts marked ●, please contact us or distributor, because they need special technical knowledge.

The following table shows the intervals of inspection and maintenance under operating conditions. Perform the inspection and maintenance, whenever the under mentioned period or operation hours come earlier.

(A) ; In case that total operation hours per one year are less than 3,000 hours

Inspection items		Inspection points	Maintenance interval						Remarks
			Daily	1 month	Every 1 year	Every 2 years	Every 4 years	Every 6 years	
Indication of each gauge			○						See the item "Daily maintenance"
Motor	Coil	Insulation resistance			●				Dry or change if less than 1M Ω.
	Bearing	Change					●		
	Sirocco fan	Check			○				Clean if too dirty.
Compressor air-end	Bearing	Change						※2●	※1●
	Mechanical seal	Check/Change			○ (Check)		● (Change)		Check for oil leak
	O-ring	Change					●		
Air-filter		Clean/Change			○ (Change)				Clean the filter whenever the warning lamp for clogging goes on.
Dustproof air filter		Clean/Change	○ (Clean)		● (Change)				
Separator receiver tank	Oil level	Check	○						In case of oil shortage, replenish it.
	Receiver tank drain	Drain	○ (Every week)						
	Lubricant	Change		○ (At 4 month for the first time)	○				Change if too dirty.
	Oil level gauge	Clean/Change				○ (Clean)	● (Change)		
	O-ring of oil filler	Check			○				Re-usable after check, if normal.
	Oil separator element	Change			○				
Pipe system	Scavenging orifice	Clean strainer		○ First time	○				
	Oil filter	Change		○ (At 4 month for the first time)	○				
	Nylon tube	Check/Change			○ (Check)	● (Change)			
	Rubber hose	Check/Change			○ (Check)		● (Change)		
	O-ring	Check/Change			○ (Check)		● (Change)		
	Check valve for scavenging mechanical seal drain	Check/Change				● (Check / Change)			Re-usable after check, if normal.
	Check valve for vacuum relief valve	Check/Change				● (Check / Change)			Re-usable after check, if normal.
Oil cooler/After cooler		Clean			●				Clean if too dirty.
Capacity controller	Consumable part of unloader	Change					●		
	O-ring of oil quantity adjustment valve	Change					●		
	Set pressure	Confirm	○						
Safety equipment	Safety valve	Check performance		○					
	Pressure control valve	Check performance					○		
		Check/Change O-ring					●		
		Check/Change piston					●		Re-usable after check, if normal.
	Cylinder discharge air temp. sensor Separator discharge air temp. sensor	Check					●		Re-usable after check, if normal.
	Bypass valve	Change Pellet					●		
Solenoid valve		Change					●		Re-usable after check, if normal.
Belt	Adjust tension			○					If abnormal, replace it.
	Change						●		Replace the pulley, depending on its conditions.

4 Periodic Inspection/Maintenance

Regarding the periodical maintenance between inspection period and operation hours, maintenance or inspection should be performed, roughly according to 250 hours/month, 3,000 hours/year, 12,000 hours/4 years, 18,000 hours/6 years and 24,000 hours/8 years.

※1 shows the interval time of bearing replacement for the compressor of the pressure of 0.69MPa.

※2 shows the interval time of bearing replacement for the compressor of the pressure of 0.83MPa.

(B) ; In case that total operation hours per one year are less than 6,000 hours

Inspection items		Inspection points	Maintenance interval							Remarks
			Daily	1 month	Every 1 year	Every 2 years	Every 4 years	Every 6 years	Every 8 years	
Indication of each gauge			○							See the item "Daily maintenance"
Motor	Coil	Insulation resistance			●					Dry or change if less than 1M Ω.
	Bearing	Change					●			
	Sirocco fan	Check			○					Clean if too dirty.
Compressor air-end	Bearing	Change						※2●	※1●	
	Mechanical seal	Check/Change			○ (Check)		● (Change)			Check for oil leak
	O-ring	Change					●			
Air-filter		Clean/Change			○ (Change)					Clean the filter whenever the warning lamp for clogging goes on.
Dustproof air filter		Clean/Change	○ (Clean)		● (Change)					
Separator receiver tank	Oil level	Check	○							In case of oil shortage, replenish it.
	Receiver tank drain	Drain	○ (Every week)							
	Lubricant	Change		○ (At 2 month for the first time)	○ (Every 6month)					Change if too dirty.
	Oil level gauge	Clean/Change				○ (Clean)	● (Change)			
	O-ring of oil filler	Check			○					Re-usable after check, if normal.
	Oil separator element	Change			○					
Pipe system	Scavenging orifice	Clean strainer		○ First time	○					
	Oil filter	Change		○ (At 2 month for the first time)	○ (Every 6month)					
	Nylon tube	Check/Change			○ (Check)	● (Change)				
	Rubber hose	Check/Change			○ (Check)		● (Change)			
	O-ring	Check/Change			○ (Check)		● (Change)			
	Check valve for scavenging mechanical seal drain	Check/Change				● (Check / Change)				Re-usable after check, if normal.
	Check valve for vacuum relief valve	Check/Change				● (Check / Change)				Re-usable after check, if normal.
Oil cooler/After cooler		Clean			●					Clean if too dirty.
Capacity controller	Consumable part of unloader	Change				●				
	O-ring of oil quantity adjustment valve	Change					●			
	Set pressure	Confirm	○							
Safety equipment	Safety valve	Check performance		○						
	Pressure control valve	Check performance				○				
		Check/Change O-ring				●				
		Check/Change piston				●				Re-usable after check, if normal.
		Cylinder discharge air temp. sensor Separator discharge air temp. sensor	Check					●		
	Bypass valve	Change Pellet					●			
Solenoid valve		Change				●				Re-usable after check, if normal.
Belt		Adjust tension			○					If abnormal, replace it.
		Change					●			Replace the pulley, depending on its conditions.

4 Periodic Inspection/Maintenance

Regarding the periodical maintenance between inspection period and operation hours, maintenance or inspection should be performed, roughly according to 500 hours/month, 6,000 hours/year, 24,000 hours/4 years, 36,000 hours/6 years and 48,000 hours/8 years.

※1 shows the interval time of bearing replacement for the compressor of the pressure of 0.69MPa.

※2 shows the interval time of bearing replacement for the compressor of the pressure of 0.83MPa.

Periodic Inspection of dryer

Items	Points	Maintenance interval			Remark
		Daily	Monthly	Yearly	
Auto drain valve	Check/Clean	○ (Check)	○ (Clean)		Make sure condensate is properly drained.
Fan motor	Check			●	Make sure motor turns smoothly and without unusual noise.
Refrigerator	Check		○		Make sure there is no abnormal noise and no oil leak.
Condenser	Clean			●	Clean dust and dirt.

Regarding the items or parts marked ○ in the table of periodical maintenance customers or users should perform periodical maintenance of the items or parts marked ○ by themselves. For such items or parts marked ●, please contact us or distributor, because they need special technical knowledge.

- Only main components are mentioned in the table.
- Maintenance interval means that periodical maintenance should be repeatedly carried out.
- Use AIRMAN genuine parts and recommended oil for replacement.
- The maintenance intervals mentioned above are not guaranteed periods.

4 Periodic Inspection/Maintenance

4.2 Inspection/Maintenance

Perform this job of inspection and maintenance safely, referring to the instruction manual for routine inspection/maintenance.

Herewith only such items which are not mentioned in the instruction manual supplied together with each unit shall be explained.

4.2.1 Electric motor

(1) Inspection of insulation resistance of motor coil

IMPORTANT

- Make sure to switch off power supply switch.
- Make sure to separate cables from contactor.
- Never touch any terminal of each cable because insulation resistance will be reduced if touched.

1. Insulation between phase

- ① Separate motor cable from contactor.
- ② Measure the insulation resistance between U and V, V and W, then W and U.
- ③ Results of measurements: In case it is more than 1Ω , normal.
In case it is less than 1Ω , abnormal.

2. Insulation between ground and motor body

- ① Separate motor cable from contactor.
- ② Measure the insulation resistance between U and motor body (motor ground), V and motor body, then W and motor body.
- ③ Results of measurements: In case it is more than 1Ω , normal.
In case it is less than 1Ω , abnormal.

3. Check coiled wires for disconnection (For reference)

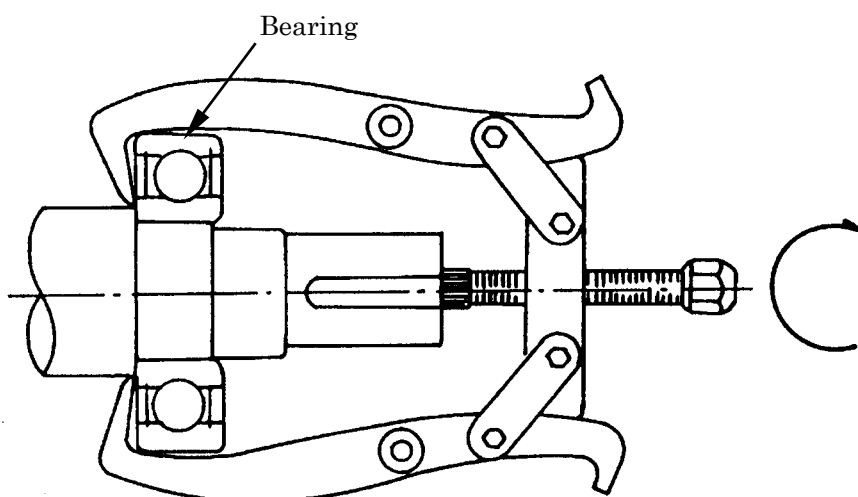
- ① Separate motor cable from contactor.
- ② Measure the coiled wires resistance between U and X, V and Y, then W and Z with insulation resistance tester.
- ③ Results of measurements: In case it is almost 0Ω , normal.
In case it is infinite (∞), abnormal (disconnection).

4 Periodic Inspection/Maintenance

(2) Replacement of motor bearing

Perform the replacement of motor bearing according to the following procedures.

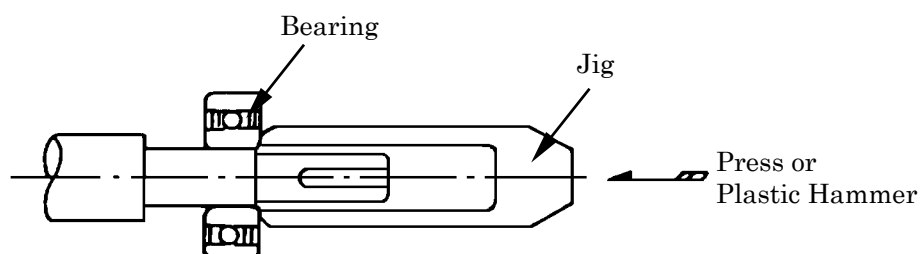
- ① Remove the fixing bolts for the brackets on both load and no-load sides, and then remove the brackets from the frame, softly patting them with plastic hammer.
- ② Then pull out the rotor from the stator, preventing it from touching the stator.
- ③ Extract the motor bearing, using such tool as gear puller shown below.



TXT079

- ④ Install the bearing with the jig or pipe applied as shown below.

When no press is available, install it by patting a jig or pipe by a plastic hammer.



TXT080

IMPORTANT

- Never hammer directly the bearing with a metal hammer for inserting the bearing.
- Never press it in by hammering holding the outer-lace of the bearing with metal pad.
- As the bearing is installed with non-contact rubber seal and heat resisting long-life grease filled, contact us for the details.
- It is impossible that the bearing is pushed in by heating.

4 Periodic Inspection/Maintenance

4.2.2 Separator receiver tank

(1) Cleaning and replacing oil level gauge

- ① When oil level gauge is so soiled that oil level cannot be checked, remove it and spray the brake cleaning agent (used for automobiles) (components: toluene 40 – 45% and hexane 25 – 30%) into the gauge several times.
- ② Periodically replace it whenever the replacement time comes.

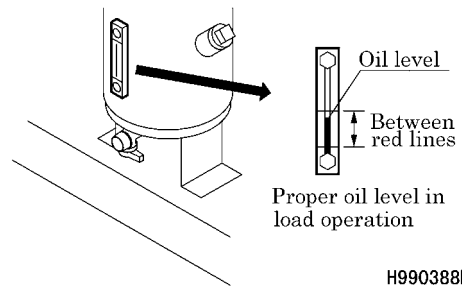
(2) How to install oil level gauge

Install the oil level gauge in the specified direction, namely with the red line downward.

Tighten the bolts for level gauge, applied by the following torques. Other torques than those specified below can cause damages and oil leakage trouble.

26.0N·m(255kgf·cm)

Compressor oil quantity is normal when the oil level is indicated between red lines of oil level gauge during operation.



4.2.3 Discharge air sensor, air temperature sensor at cylinder outlet, and air temperature sensor at separator outlet

(1) Checking the function

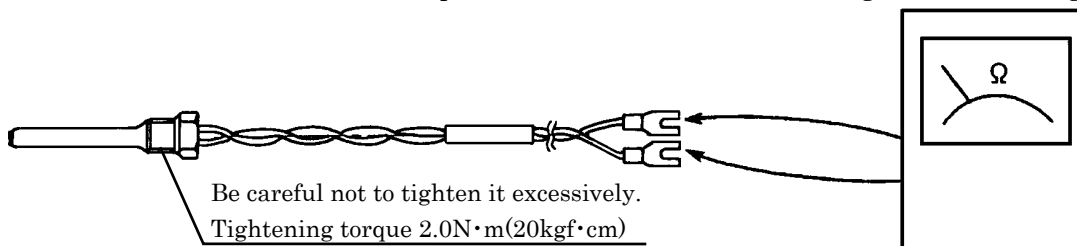
Check the function according to the following procedures.

- ① During operation, remove the sensor cable connected to the controller.
- ② Make sure to confirm that the unit is stopping and the lamp indicating abnormality glows.

(2) Checking the set temperature

Checking the set temperature according to the following procedures.

- ① Separate the cable between controller and sensor.
- ② Confirm the relations between temperature and resistance according to the following table.



Features of temperature and resistance

Temperature °C	Features of criteria resistance (Ω)	Range of Resistance (Ω)
10	18028.8	19905.0~16314.7
15	14416.4	15845.7~13104.2
20	11604.7	12700.0~10594.3
25	9401.0	10254.1~8618.8
80	1318.2	1256.9~1381.2
105	640.1	619.3~661.1
110	559.8	543.0~576.5

S_M002

4 Periodic Inspection/Maintenance

4.2.4 By-pass valve

(1) Principle of function

The valve is provided with pellets. The pellets shrink according to the temperature of flowing oil of the valve so that the oil flowing passage can be automatically changed. When the oil temperature is low, the pellet is kept shrunken so that the oil flows through the by-pass passage of oil filter. As the temperature rises, the pellet expands to close the oil passage to the by-pass passage and to open the flow outlet to oil cooler. Therefore, the oil flows to the oil cooler, thus preventing the oil temperature from abnormally rising.

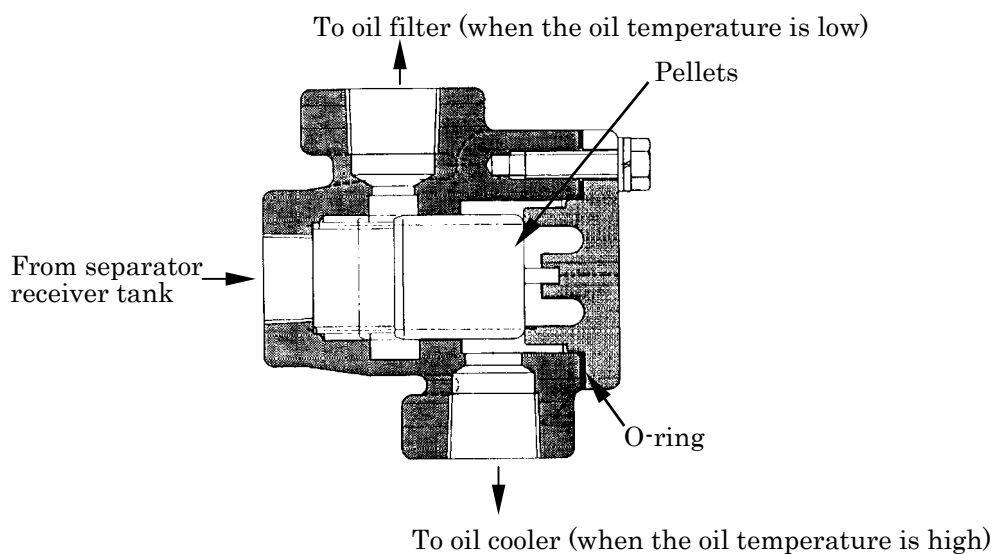
(2) Checking function

The by-pass valve functions correctly in case any abnormal rise of temperature and any other abnormalities are not found.

(3) Replacement of the pellets

Remove the cover and replace the pellets and O-ring. Coat the new O-ring with compressor oil and then tighten the cover.

*The following Fig. shows the sectional view of By-pass valve when the oil temperature is low.



S_M003

5. Electrical System

5.1 Controller

5.1.1 Specification

Item		Contents	
Function of Controller	Operation mode	• Continuous operation (Pressure range automatic adjusting system) • E Mode operation (Function of most suitable pressure operation) • Auto start/stop (Stop judgment time adjusting system) • Remote control • Dryer advance operation • Instantaneous re-start • Multi-units control operation (Option)	
	Indicator lamps on operation panel	Digital indication	• Operation time
			• Current
		Indicator lamp	• Start • Stop • ECO lamp
			• Air filter • Dryer • Cylinder discharge air temp. (Lamp flickers at 105°C) • Separator discharge air temp. (Lamp flickers at 105°C)
			• Cylinder discharge air temp. (Lamp glows at 110°C) • Separator discharge air temp. (Lamp glows at 110°C) • Over current • Opposite phase • Pressure sensor
	Remote side indicator lamp	• Operating • Abnormal • Remote selection signal	
	Emergency stop (Start lock)	• Cylinder discharge air temp. sensor disconnected or discharge air temp. rises up to 110°C • Separator discharge air temp. sensor disconnected or discharge air temp. rises up to 110°C • When thermal relay functions • Power supply is connected in reverse phase and T phase is missing • Pressure sensor disconnected or discharge pressure exceeds 0.98MPa (10kgf/cm²)	
	Adjusting range of discharge air pressure	• Adjusting range of full load recovery pressure : 0.50~0.79MPa (5.1~8.0kgf/cm²) • Adjusting range of unload starting pressure : differential pressure against full load recovery pressure 0.08~0.39MPa (0.8~4.0kgf/cm²)	
	Judgment time for auto start/stop mode stopping	• 30~150 seconds	
	Y-Δ timing	• Y operation time 4 ± 0.2 seconds	
		• Y-Δ selection time 0.048 ± 0.008 seconds	
	Starting unloader	• 5 ± 0.2 seconds	

5. Electrical System

*Function of control output relay

Symbol	Function	Symbol	Function
RY0	• It switches ON by operation This is used as operation indicator lamp for remote control operation	RY5	• Capacity control signal Switching it ON, it starts full load operation. Switching it OFF, it starts unload operation
RY1	• Dryer operation signal Switching it ON, it starts dryer operation. Switching it OFF, it stops	RY6	• Failure signal When it is in trouble, it switches ON. When warning occurs, it will not output
RY2	• It switches ON by operation	RY7	• Remote selection signal It switches ON by remote selection
RY3	• It switches ON by Y operation		
RY4	• It switches ON by Δ operation		

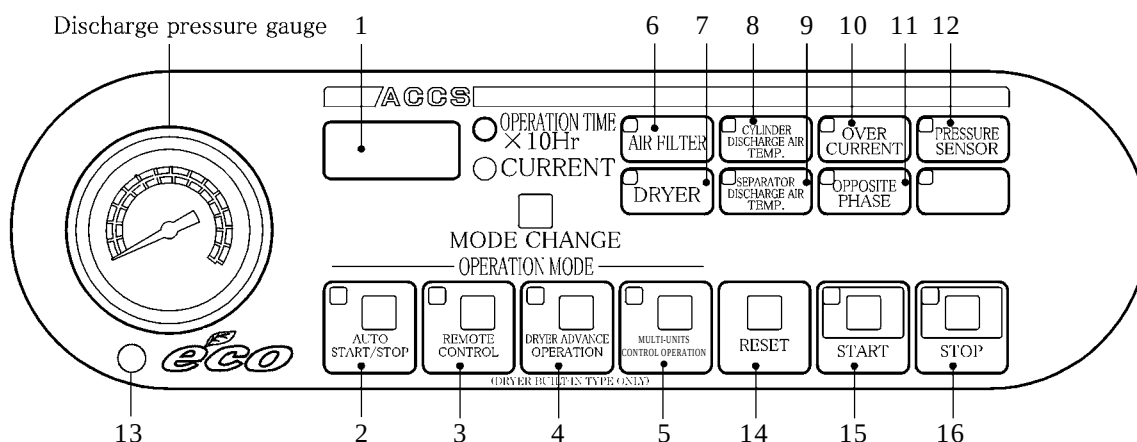
5.1.2 Explanation about operation modes

Operation mode		Contents																						
Continuous operation		When air consumption decreases and discharge pressure rises, unloader valve functions to close the inlet port to compressor air end and accordingly oil feed adjusting device functions to minimize oil feed. At the same time, it collects oil and compressed air from the discharge port of air end and sends them into the separator receiver tank, thus reducing the power of compressor.																						
E Mode operation		When selecting E mode by E mode selection switch, the set pressure is automatically switched to the low pressure operation with 0.04MPa (0.4kgf/cm ³), thus enabling energy saving operation.																						
Auto start/stop operation		Conditions of Auto Stop •When operation time passes 10 minutes or more, and unload integrating time exceeds 60 seconds, and further longer stop time than stop judging time is anticipated. •After dryer built-in type unit stops, dryer continues to operate. Conditions of Re-start In case the pressure on load drops till full load recovery pressure. But within 30 seconds after stopping, it will not re-start because this time is required for discharging compressed air.																						
Remote control operation		Wired to the remote control terminal plate, start/stop operation can be done remotely. Stop operation can be done at machine side also. <table border="1"><tr><td rowspan="4">Remote control button</td><td colspan="2">Machine side</td><td colspan="2">Remote control</td></tr><tr><td>Start</td><td>Stop</td><td>Start</td><td>Stop</td></tr><tr><td>OFF</td><td>○</td><td>○</td><td>×</td><td>×</td></tr><tr><td>ON</td><td>×</td><td>○ ※1</td><td>○ ※2</td><td>○</td></tr></table> ※1 If stopped at machine side, remote control mode is reset. In case remote control mode is selected again, button ON after pushing OFF first. ※2 Remote start can be done while remote start contact is OFF → (ON continuous). Therefore, while remote start contact is ON, it will not start even after pushing ON microprocessor remote start button.				Remote control button	Machine side		Remote control		Start	Stop	Start	Stop	OFF	○	○	×	×	ON	×	○ ※1	○ ※2	○
Remote control button	Machine side		Remote control																					
	Start	Stop	Start	Stop																				
	OFF	○	○	×	×																			
	ON	×	○ ※1	○ ※2	○																			
Dryer advance operation		Pushing start button, dryer starts first, and then unit start 3 minutes later.																						
Instantaneous stop re-start operation		Even when compressor stops by instantaneous stop of power, and if it will be recovered within 2 seconds, it begins to run again 30 seconds after recovery. The operation mode before stopping is maintained even after restarting.																						
Multi-units control operation (Option)		When operating some units in parallel, automatic control of the units in accordance with the air consumption will serve to save much energy. This is a rotary multi-unit control system. Starting the unit which is operating for the shortest time, and stopping the unit which is operating for the longest time. Thus it can equalize operation time for the units which are operating.																						

For the details, see instruction manual.

5. Electrical System

5.1.3 Operation panel



A050467E

Item	Contents
1 Digital monitor	- When switching power ON, this monitor digitally indicates operating hour $\times 10$ hr. - Pushing the selector switch, it digitally displays motor current at A. - In order to make this monitor display the operating hour again, push the selection switch ON.
2 Auto start/stop button	- During auto stop (both auto start/stop indicator and start indicator lamps light on), pushing auto start/stop button, auto start/stop mode will be released and compressor will stop. (only stop indicator lamp lights) - During auto stop (both auto start/stop indicator and start indicator lamps light on), pushing the stop button, auto start/stop mode will be maintained, but compressor stops and it will not restart. (both auto start/stop indicator and stop indicator lamps light on)
3 Remote control button	- While remote control indicator lamp glows, it is possible only to start remotely. - It is possible to stop operation at machine side and remotely too. But when stopping operation at machine side, remote control mode is reset and model become standard mode. When selecting remote control mode again, button OFF the remote control button and extinguish remote control indicator, then push remote control button and turn on remote control indicator again. - It is possible to change standard mode to remote control mode during machine moving under standard mode, if remote side connection is ON condition. - In case the remote side starting contact is "ON", it will not start even by to the remote control mode. First, switch the remote control mode "OFF" and then switch it "ON". Then it can be started.
4 Dryer advance operation button	- Even when dryer is not connected or mounted, the indicator lamp of dryer advance operation glows with this button pressed. - When light on the dryer advance indicator lamp, push start button. Compressor will start after 3 minutes, even with dryer not connected. - When dryer advance indicator lamp lights, dryer will stop and compressor will not start if stopping dryer advance operation (namely, pushing the dryer advance operation button again) before starting compressor, dryer stops and compressor will not start. - In case of auto start/stop operation, even if compressor automatically stops, dryer still continue operating.

5. Electrical System

Item		Contents
5	Multi-units control operation button (Option)	•The number of the units to be controlled by this system: Any number of units can be controlled with this system without limit. •It is possible to intently decide the first starting unit by pushing the start button of the first unit. Thus multi-unit control system can be performed. •It is possible to skip such unit which is troubled or any unit which is not controlled by this system, namely possible to exclude such units from the units controlled by this system. •It is possible to keep the first starting unit operated under this system and the other units operated at full load. Thus it prevents useless start/stop operation, and consequently it enables most efficient operation.
6	Air filter indicator lamp	• When air filter gets clogged and differential pressure reaches 6.2kPa (635mAq), lamp glows after 10 seconds. •While operating compressor, lamp glows 10 seconds later after the cable to the differential pressure switch gets short circuited.
7	Dryer indicator lamp	•Lamp will glows 1 second after each sensor of dryer functions or unit stop by thermal relay function. •When it stops due to disconnection of dryer, it will not glow.
8	Cylinder discharge air temp. indicator lamp	•When air temp. reaches 105°C, lamp flickers. •When air temp. reaches 110°C, it causes emergency stop. •Lamp glows due to disconnection or short circuit of thermo sensor and it causes emergency stop.
9	Separator discharge air temp. indicator lamp	•When air temp. reaches 105°C, lamp flickers. •When air temp. reaches 110°C, it causes emergency stop. •Lamp glows due to disconnection or short circuit of thermo sensor and it causes emergency stop.
10	Over current indicator lamp	•When thermal relay functions, lamp glows and it causes emergency stop 0.4 seconds later. •When thermal relay functions and it stops it is impossible to release it to push reset button, if thermal relay is set manual operation.
11	Opposite phase indicator lamp	•In case of reverse phase connection or T phase missing, lamp glows 1 second after power is supplied, and it is impossible to start. When R or S phase is missing, monitor will not indicate because power supply for controller is not sent. And it can not start. •If phase missing occurs while compressor is running, it stops. (Only in case that phase loss occurs to T phase, the indicator lights on.) •It is possible to detect reverse phase connection of power supply, but it is impossible to detect reverse phase connection of motor. If motor only is connected in reverse phase, the motor starts to turn in reverse direction.
12	Pressure sensor indicator lamp	•When pressure sensor is disconnected or discharge pressure exceeds 0.98MPa (10kgf/cm²), the indicator lights on and compressor stops make emergency stop.
13	ECO lamp	•This lamp lights on during unloaded operation.
14	Reset button	•Unless contents of warning and abnormalities are revised, it is impossible to release the mode. • Even if they are revised, it is impossible to extinguish warning and abnormality lamps unless they are reset.
15	Start button/ Indicator lamp	•After power is supplied, it starts by pushing this button. •Lamp glows during machine moving. During automatic stop under auto start/stop mode, and also during operation under dryer advance operation mode, indicator lamp glows. •Indicator lamp flickers while waiting for instantaneous restart. •Even if the start button is pushed on while compressor is kept stopping by the auto start/stop mode, it will not start again unless the receiver tank pressure drops.
16	Stop button/ Indicator lamp	•Pushing stop button during remote control operation, it will stop. •When power is supplied, indicator lamp glows. During operation lamp is OFF. •During automatic stop under auto start/stop mode, indicator is OFF. •During stop under auto start/stop mode, (both auto start/stop indicator and start indicator lamps light on), pushing the stop button, auto start/stop mode will be maintained, but compressor stops and it will not restart. (Both auto start/stop indicator and stop indicator lamps light on)

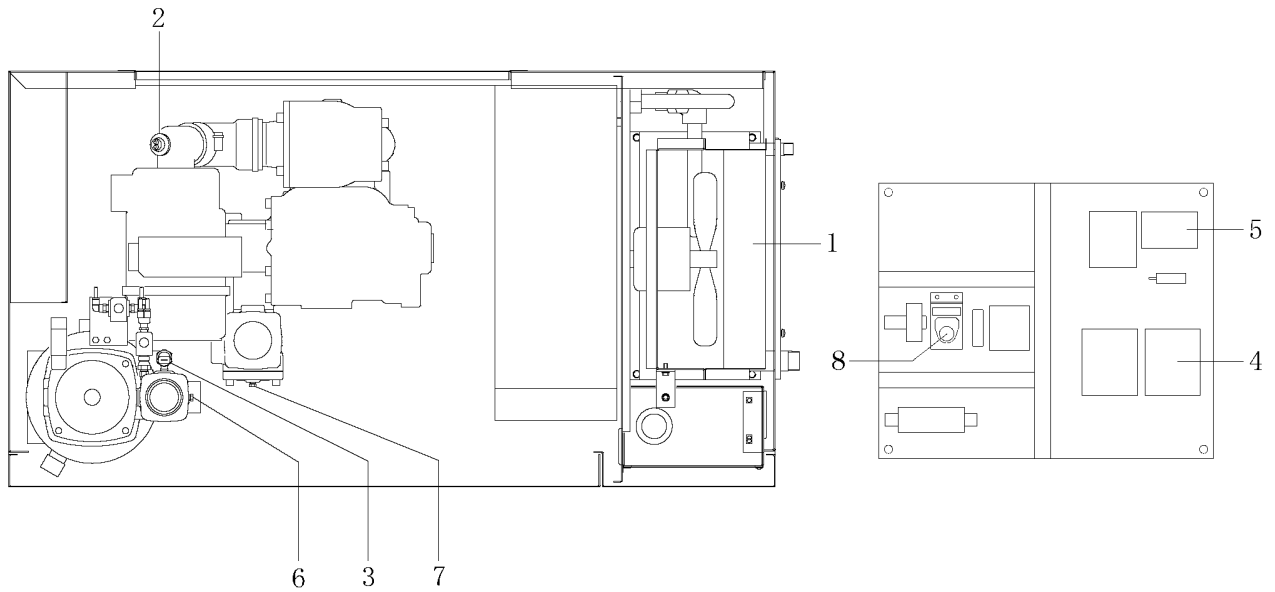
5. Electrical System

5.1.4 Checking indication of warning & abnormalities and also checking emergency stop

	Item	How to check	Normal function	Time lag
Warning	Air filter	Short circuit the cable connected to the air filter differential pressure switch.	Warning indicator lamp glows.	10 seconds
	Cylinder discharge air temp.	Remove the sensor from controller and load resistance of 619.3~661.1 Ω to controller with variable resistor.	Warning indicator lamp flickers.	—
	Separator discharge air temp.	Remove the sensor from controller and load resistance of 619.3~661.1 Ω to controller with variable resistor.	Warning indicator lamp flickers.	—
	Dryer	During operation, trip thermal relay by hand. (Be careful about electric shock.)	Warning indicator lamp glows.	1 second
Abnormality/Emergency Stop	Cylinder discharge air temp.	- Remove sensor from controller during operation. - Remove sensor from controller and load 543.0~576.5 Ω of resistance to controller by variable resistor.	Abnormality indicator lamp glows and unit stops.	—
	Separator discharge air temp.	- Remove sensor from controller during operation. - Remove sensor from controller and load 543.0~576.5 Ω of resistance to controller by variable resistor.	Abnormality indicator lamp glows and unit stops.	—
	Over current	During operation thermal is tripped by hand.	Abnormality indicator lamp glows and unit stops.	0.4 seconds
	Opposite phase	After cutting power supply, charge R phase for T phase. Cutting power supply, remove T phase.	When power is supplied, abnormality indicator lamp glows, and unit will not start even by pushing start button.	1 second
	Pressure sensor	Remove connector of pressure sensor during operation.	Abnormality indicator lamp glows and unit stops.	—

5. Electrical System

SAS22UD-58/68

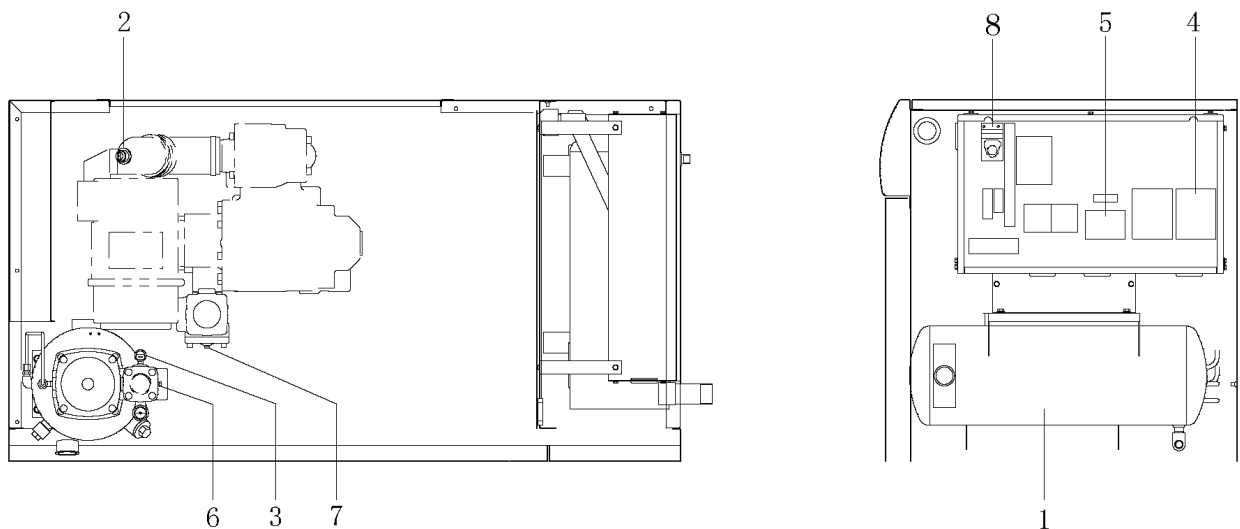


Upper side drawing

Inside of starter

S_M166

SAS37UD-58/68



Upper side drawing

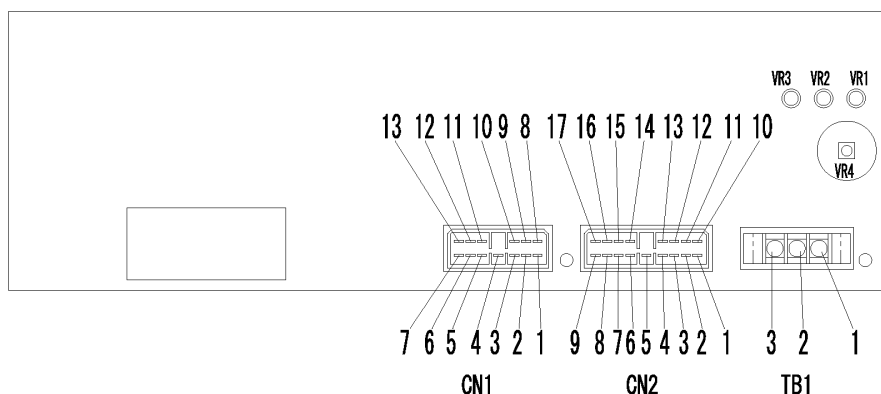
Inside of starter

S_M167

- | | |
|--|---------------------------------------|
| 1. Dryer | 5. Main motor thermal relay |
| 2. Air filter differential pressure switch | 6. Thermistor high sensitivity sensor |
| 3. Pressure sensor | 7. Thermistor high sensitivity sensor |
| 4. Connection of power supply | 8. E Mode selection switch |

5. Electrical System

5.1.5 Position of connector pins and circuit voltage



S_M105

		Item	Circuit voltage (V)	Remark
For power supply	CN1	1 Remote control selection single indicator lamp	AC100~220	Supplied by customer
		2 Power supply for remote control indicator lamp	AC100~220	Supplied by customer
		3 Remote operation indicator lamp	AC100~220	Supplied by customer
		4 Solenoid valve for capacity control	AC200/200/220	
		5 Electromagnet contactor for star	AC200/200/220	
		6 Power R2(for micon)	AC200/200/220	
		7 Power S1(for micon)	AC200/200/220	
		8 Electromagnet contactor of dryer	AC200/200/220	
		9 Remote abnormality indicator lamp	AC100~220	Supplied by customer
		10 Power R2	AC200/200/220	
		11 Electromagnet contactor for delta	AC200/200/220	
		12 Contactor for insulation	AC200/200/220	
		13 Power T1	AC200/200/220	

5. Electrical System

Item		Circuit voltage	Remark
For sensor	1	Pressure sensor output	DC0.5~4.5V(between CN2-2) When pressure 0MPa : 0.5V When pressure 0.98MPa : 4.5V
	2	Pressure sensor GND	—
	3	Current sensor	—
	4	Current sensor	—
	5	Failure of dryer	Normal : 0V Abnormal : DC12V(between CN2-11) Abnormal : OFF B connect Time lag : 1 second
	6	Air filter differential pressure switch	Normal : DC12V Abnormal : 0V(Between CN2-11) Abnormal : OFF A connect Time lag : 10 seconds
	7	Switch GND	—
	8	Remote start switch	During operation : 0V During stop condition : DC12V(Between CN2-9) Supplied by customer
	9	Remote start switch GND	—
	10	Pressure sensor power	DC5V(Between CN2-2)
	11	Switch GND	—
	12	Switch GND	—
	13	Thermal relay	Normal : 0V Abnormal : DC12V(between CN2-11) Abnormal : OFF B connect Time lag : 0.4 seconds
	14	Switch GND	—
	15	Blank	—
	16	Switch GND	—
	17	Selection switch of usual unit and standby unit	Selection of usual unit : 0V Selection standby unit : DC12V (between CN2-9) Supplied by customer
TB1	1	Separator discharge air temp. sensor	Less than DC5V(between TB1-2) Detect abnormality Resistance : 543.0~576.5Ω
	2	Air temp. sensor GND	—
	3	Cylinder discharge air temp. sensor	Less than DC5V(between TB1-2) Detect abnormality Resistance : 543.0~576.5Ω

5. Electrical System

5.2 Electrical Parts

5.2.1 Specifications

(1) Starter, Solenoid valve, Hour-meter

SAS22UD-58/68

Symbol	Name	Maker	Model	Remarks
42,52	Electromagnet contactor	MITSUBISHI Electric	S-N65	For insulation, for delta operation (surge damping valve built-in)
6	Electromagnet contactor	MITSUBISHI Electric	S-N35	For star operation (surge damping valve built-in)
52C	Electromagnet switch	Fuji Electric	FW-0	For dryer
49	Thermal relay	MITSUBISHI Electric	TH-N60KP	For protection of motor
F	Fuse holder	Sakazume	FHC-15	(AC250V,15A)
	Fuse	Not specified	$\phi 6.4 \times 30$	(AC250V,5A)
TB10	Terminal plate	Izumi	BN15MW(12P)	For remote control
SV1	Solenoid valve(3 way)	CKD	AG4X7908	For capacity control
SV2,SV3	Solenoid valve(2 way)	CKD	FAB42-X0679	For auto-relief, for separator recovery

SAS37UD-58/68

Symbol	Name	Maker	Model	Remarks
42,52	Electromagnet contactor	MITSUBISHI Electric	S-N95	For insulation, for delta operation (surge damping valve built-in)
6	Electromagnet contactor	MITSUBISHI Electric	S-N65	For star operation (surge damping valve built-in)
52C	Electromagnet switch	Fuji Electric	FW-0 RC=9A	For dryer refrigerator
52CF	Electromagnet switch	Fuji Electric	FW-0 RC=0.64A	Fan for dryer condenser
49	Thermal relay	MITSUBISHI Electric	TH-N120KP	For protection of motor
F	Fuse holder	Sakazume	FHC-15	(AC250V,15A)
	Fuse	Not specified	$\phi 6.4 \times 30$	(AC250V,5A)
TB10	Terminal plate	Izumi	BN15MW(12P)	For remote control
SV1	Solenoid valve(3 way)	CKD	AG4X7908	For capacity control
SV2,SV3	Solenoid valve(2 way)	CKD	FAB42-X0679	For auto-relief, for separator recovery

5. Electrical System

(2) Sensor, Switch

Name	Symbol	Monitor	Maker	Remarks
Air filter differential pressure switch	ACR	Air filter	NIPPON DONALDSON, LTD.	6.2kPa(635mmAg) DC12V A contact
Thermo sensor	TLE1	Cylinder discharge air temp.	SAGINOMIYA	543.0~576.5Ω at 110°C
	TLE2	Separator discharge air temp.	SAGINOMIYA	543.0~576.5Ω at 110°C
Pressure sensor	PRS	Pressure sensor	Nihon DENSO	0.5~4.5V (0~0.98MPa)
Detecting opposite phase	—	Opposite phase	—	Detect it with controller

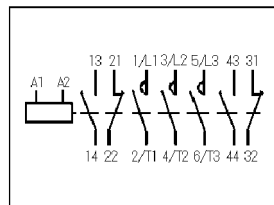
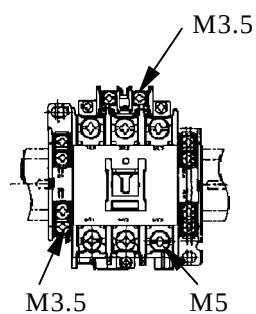
5.2.2 Electromagnet contactor

Model	Kinds		Maker	Electromagnet switch Type	Electromagnet contactor Type	Thermal relay Type	Thermal relay set value (A)
SAS22UD	For motor	For insulation	MITSUBISHI Electric	—	S-N65 2a2b	—	—
		For protection	MITSUBISHI Electric	—	—	TH-N60KP RC=54A(43~65A)	54.0/51.0/48.0
		For Star	MITSUBISHI Electric	—	S-N35 2a2b	—	—
		For Delta	MITSUBISHI Electric	—	S-N65 2a2b	—	—
	For dryer		Fuji Electric	FW-0	FC-0 1a	TR-0 RC=9A(7~11A)	9.0/9.0/9.0
SAS37UD	For motor	For insulation	MITSUBISHI Electric	—	S-N95 2a2b	—	—
		For protection	MITSUBISHI Electric	—	—	TH-N120KP RC=82A(65~100A)	90.0/85.0/80.0
		For Star	MITSUBISHI Electric	—	S-N65 2a2b	—	—
		For Delta	MITSUBISHI Electric	—	S-N95 2a2b	—	—
	For dryer refrigerator		Fuji Electric	FW-0	FC-0 1a	TR-0 RC=9A(9~13A)	10.5/10.5/10.5
	Fan for dryer condenser		Fuji Electric	FW-0	FC-0 1a	TR-0 RC=0.64A (0.64~0.96A)	0.7/0.7/0.7

5. Electrical System

(1) Electromagnet contactor

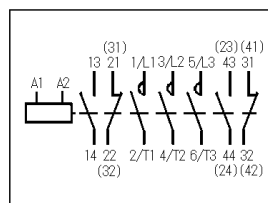
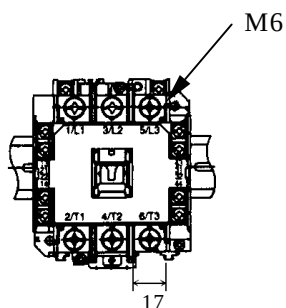
Type: S-N35



Structure of contacts

S_M160

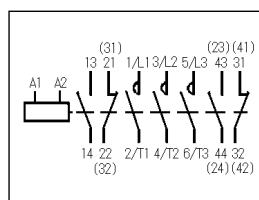
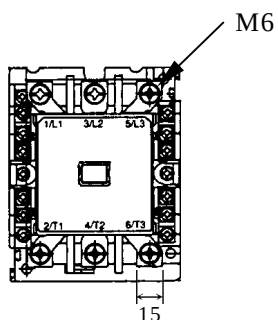
Type: S-N65



Structure of contacts

S_M094

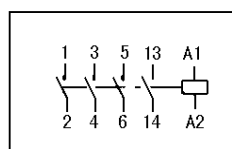
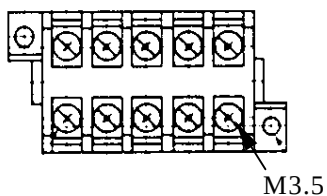
Type: S-N95



Structure of contacts

S_M095

Type: FC-0



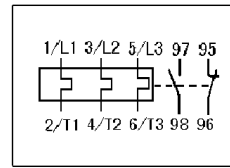
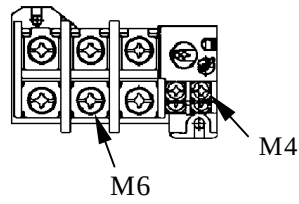
Structure of contacts

S_M093

5. Electrical System

(2) Thermal relay

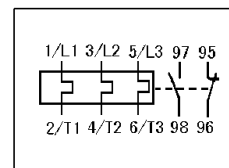
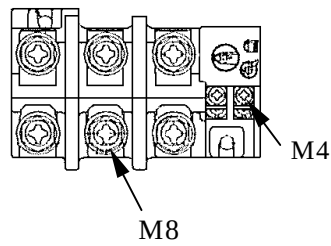
Type: TH-N60KP



Structure of contacts

S_M161

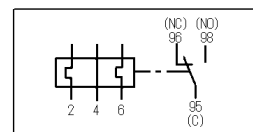
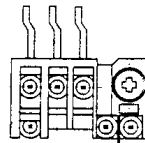
Type: TH-N120KP



Structure of contacts

S_M138

Type: TR-0



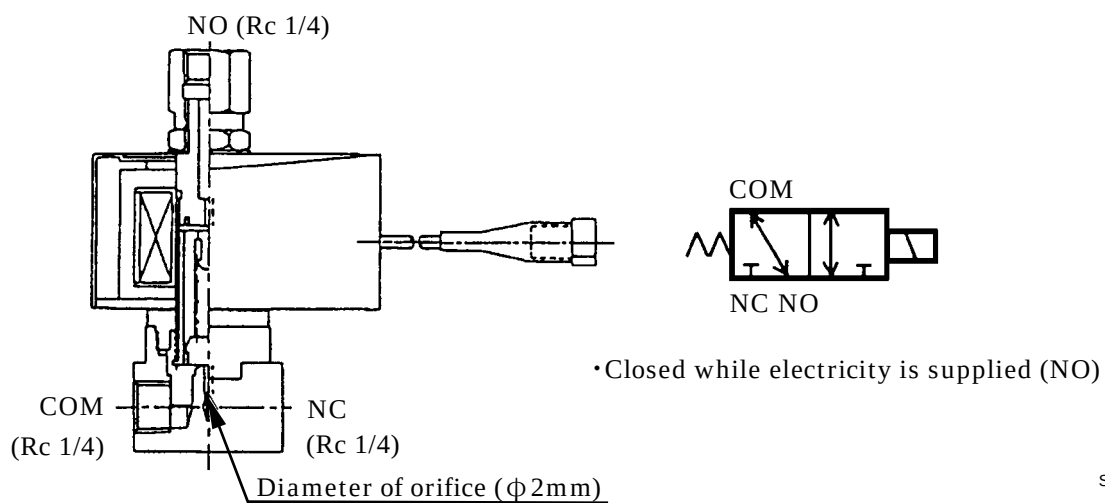
Structure of contacts

S_M025

5. Electrical System

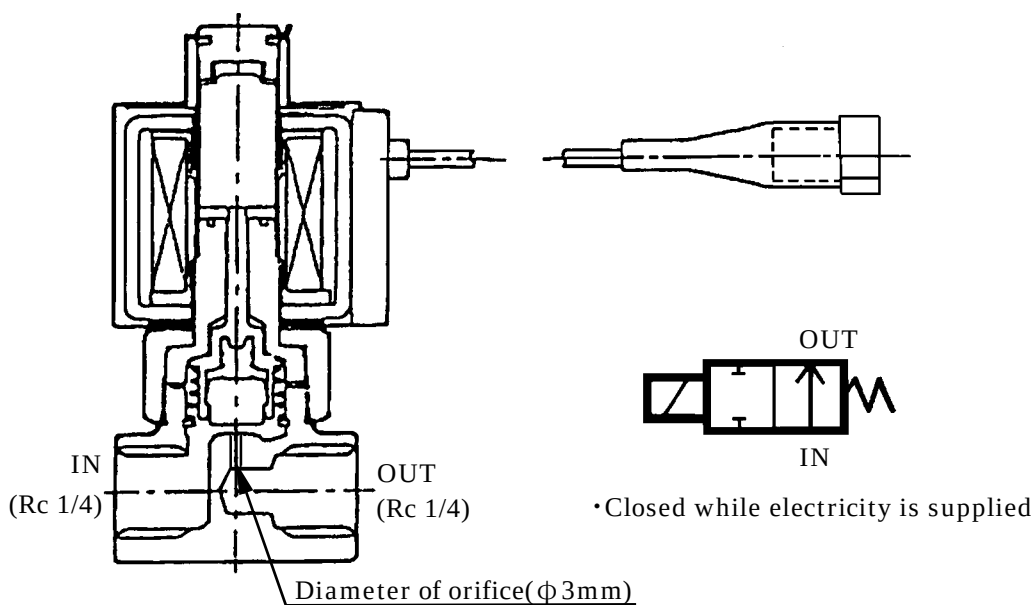
5.2.3 Solenoid valve

(1) For capacity control



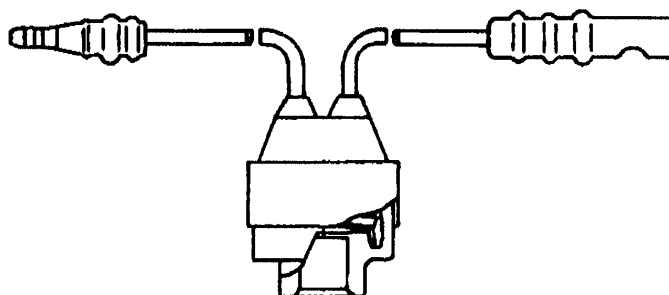
S_M120

(2) For auto-relief, for separator recovery



S_M039

5.2.4 Differential pressure switch of air filter

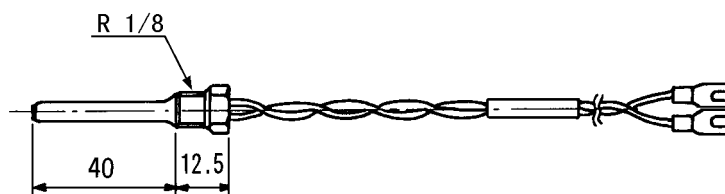


- Operating resistance: $635 \pm 58\text{mmAq}$ (initial feature)
- A Contact

TX015

5. Electrical System

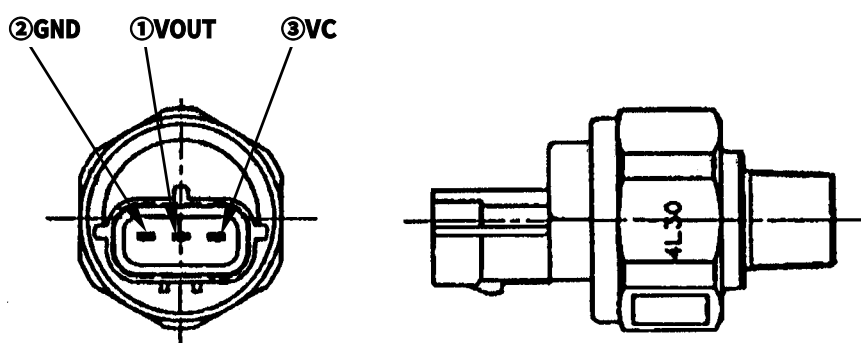
5.2.5 Thermo sensor



S_M008

※ For the relation between temperature and resistance, refer to 4.2.3.

5.2.6 Pressure sensor



SCTXT027

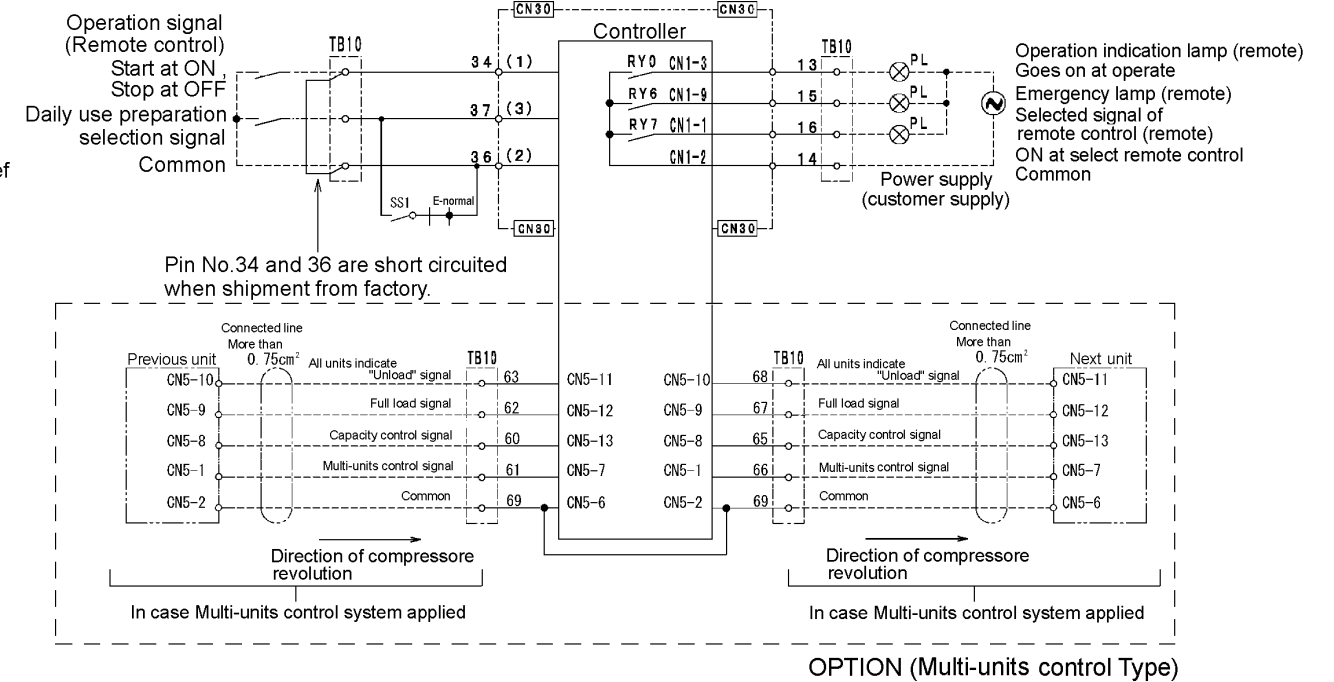
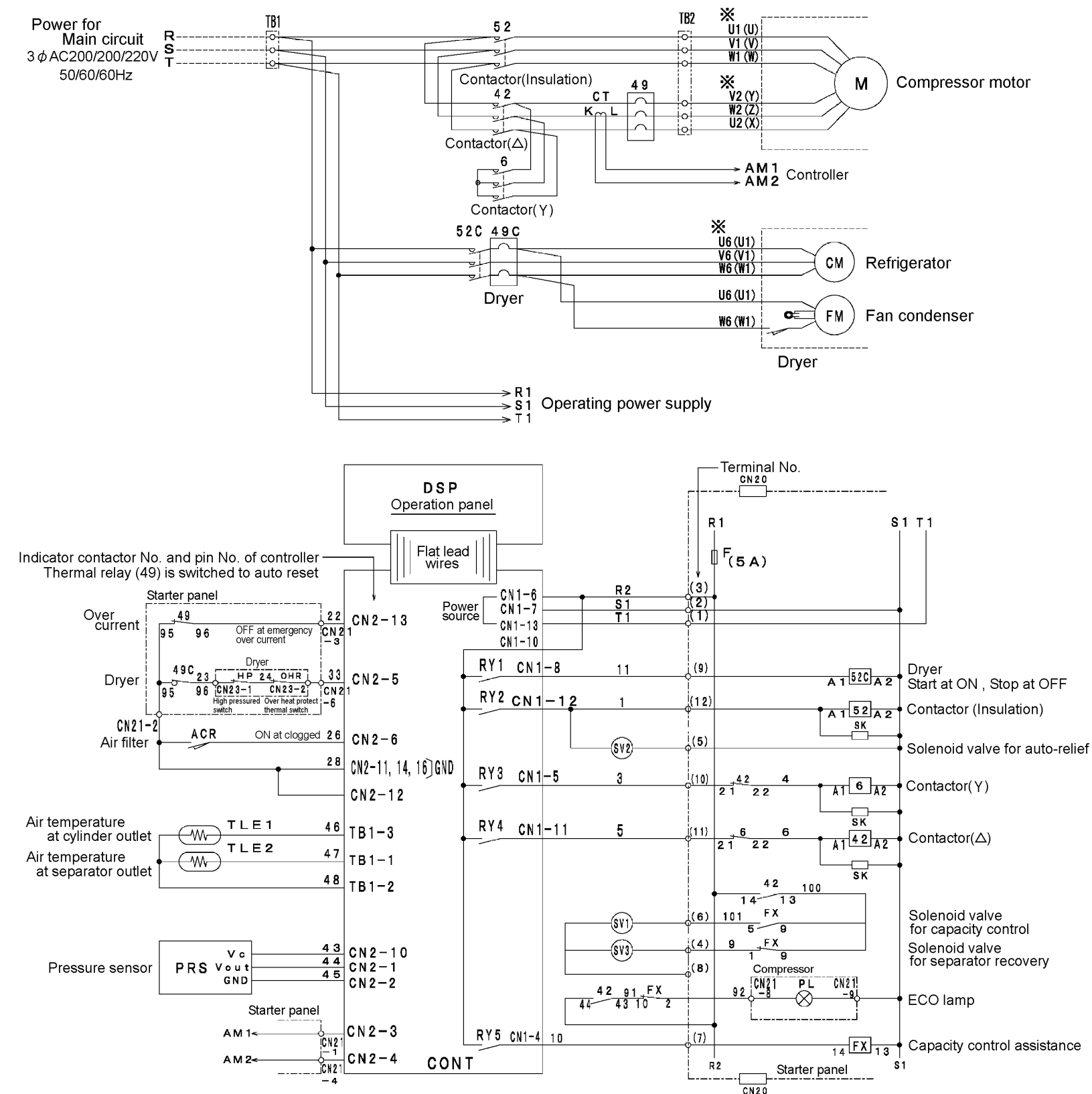
Cable	Signal
①VOUT	Outlet (DC0.5~4.5V) At the pressure 0MPa(0kgf/cm ²), DC0.5V At the pressure 0.1MPa(1kgf/cm ²), DC0.9V At the pressure 0.6MPa(6kgf/cm ²), DC2.9V At the pressure 0.69MPa(7kgf/cm ²), DC3.3V At the pressure 0.98MPa(10kgf/cm ²), DC4.5V
②GND	Common
③VC	5V

5. Electrical System

5.3 Electrical Wiring Diagram

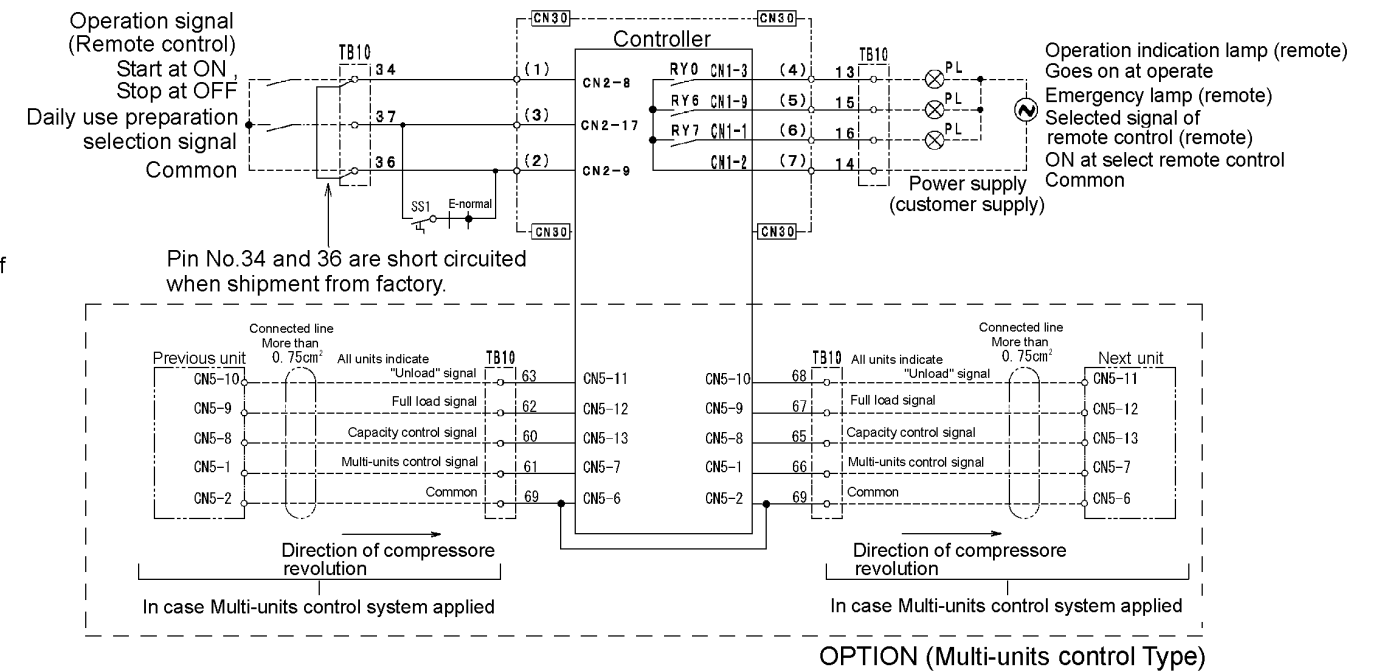
5.3.1 Dryer built-in type

(1) SAS22UD-58/68 (200/200/220V Type)



※: Such things in the parenthesis are used for the products which were produced before September, 2007.

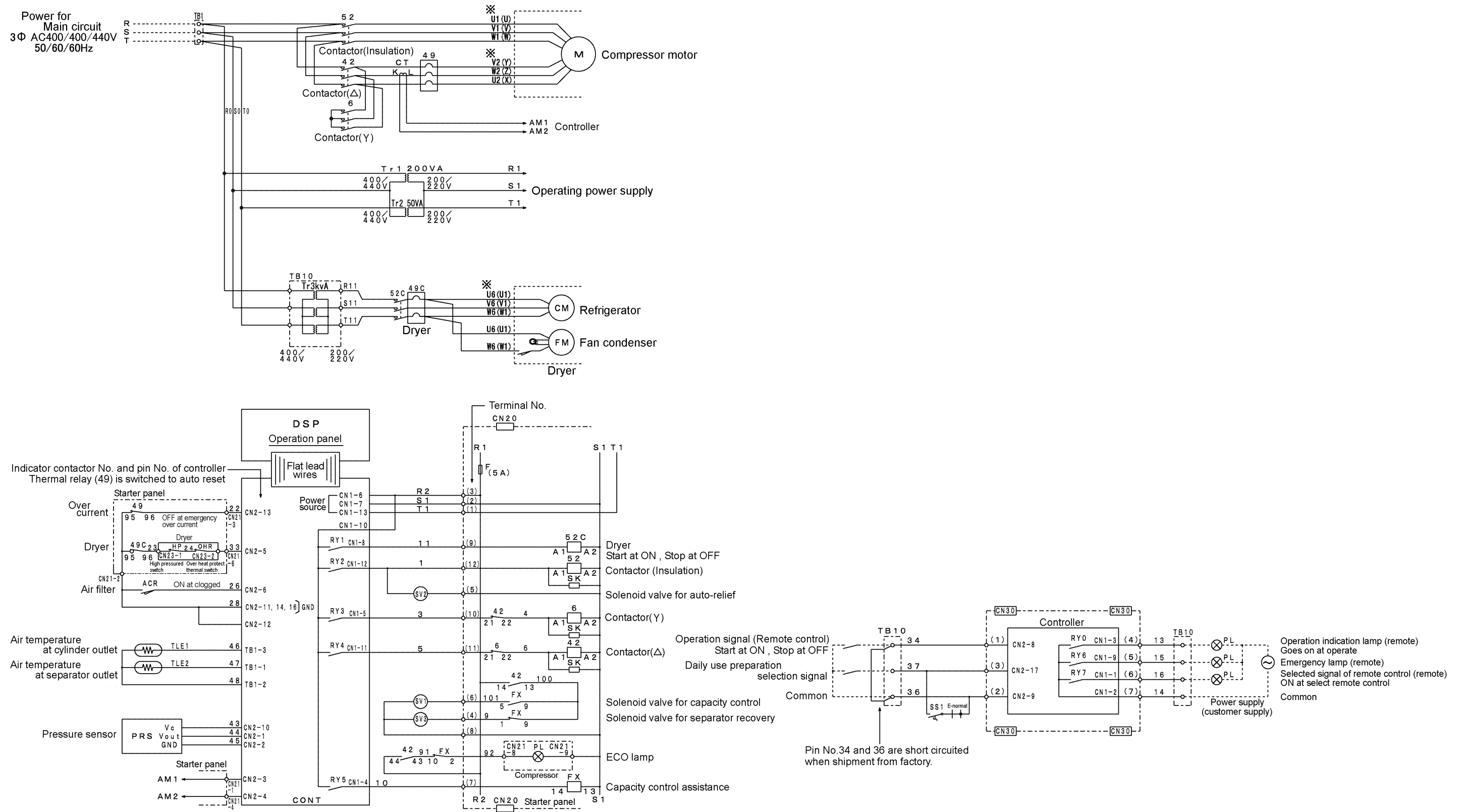
(2) SAS37UD-58/68 (200/200/220V Type)



_M169E

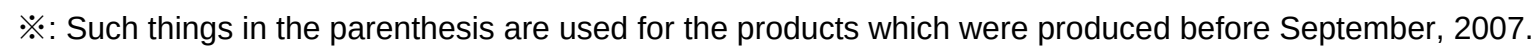
5. Electrical System

(3) SAS22UD-58/68 (400/400/440V Type)



※: Such things in the parenthesis are used for the products which were produced before September, 2007.

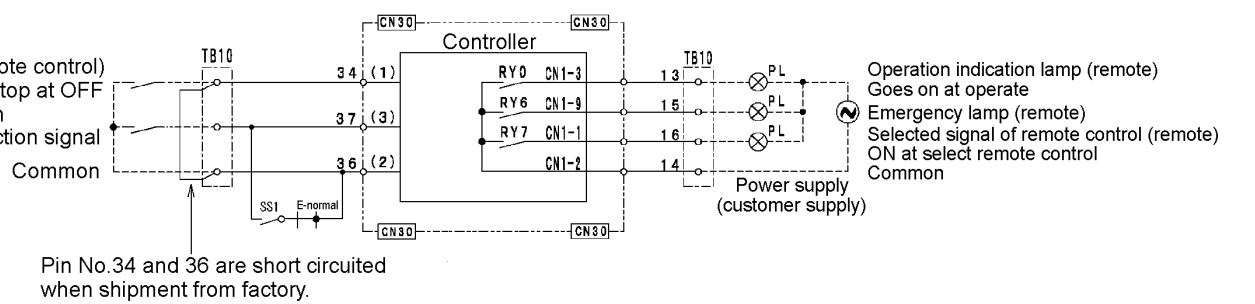
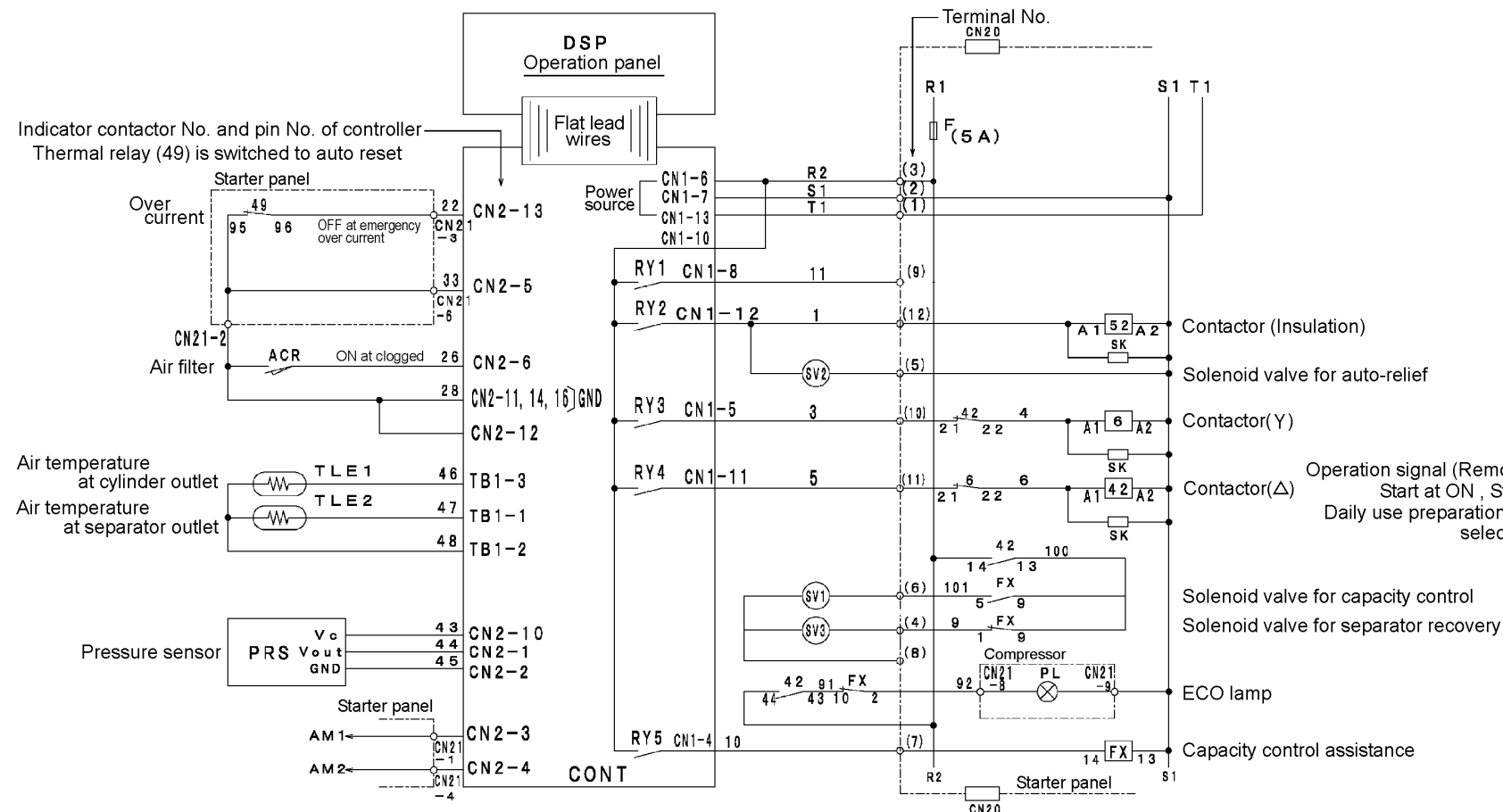
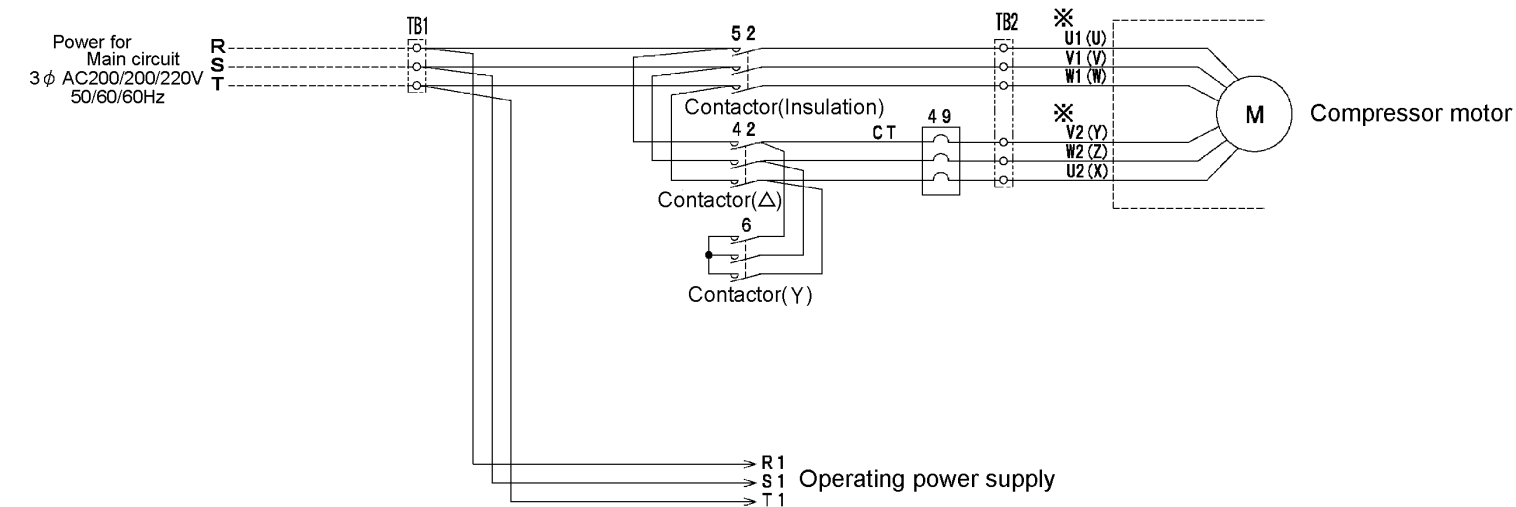
(4) SAS37UD-58/68 (400/400/440V Type)



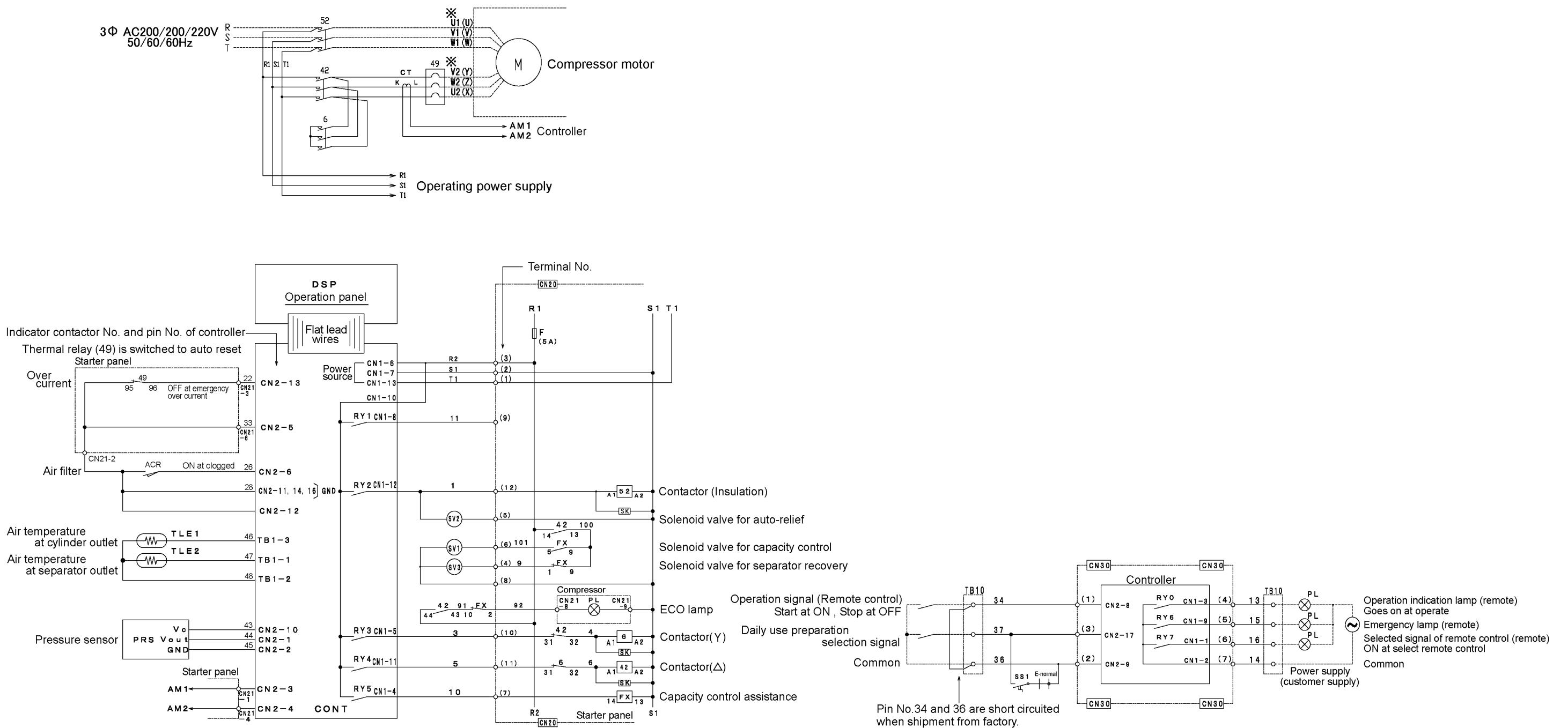
5. Electrical System

5.3.2 Without dryer type

(1) SAS22U-58/68 (200/200/220V Type)



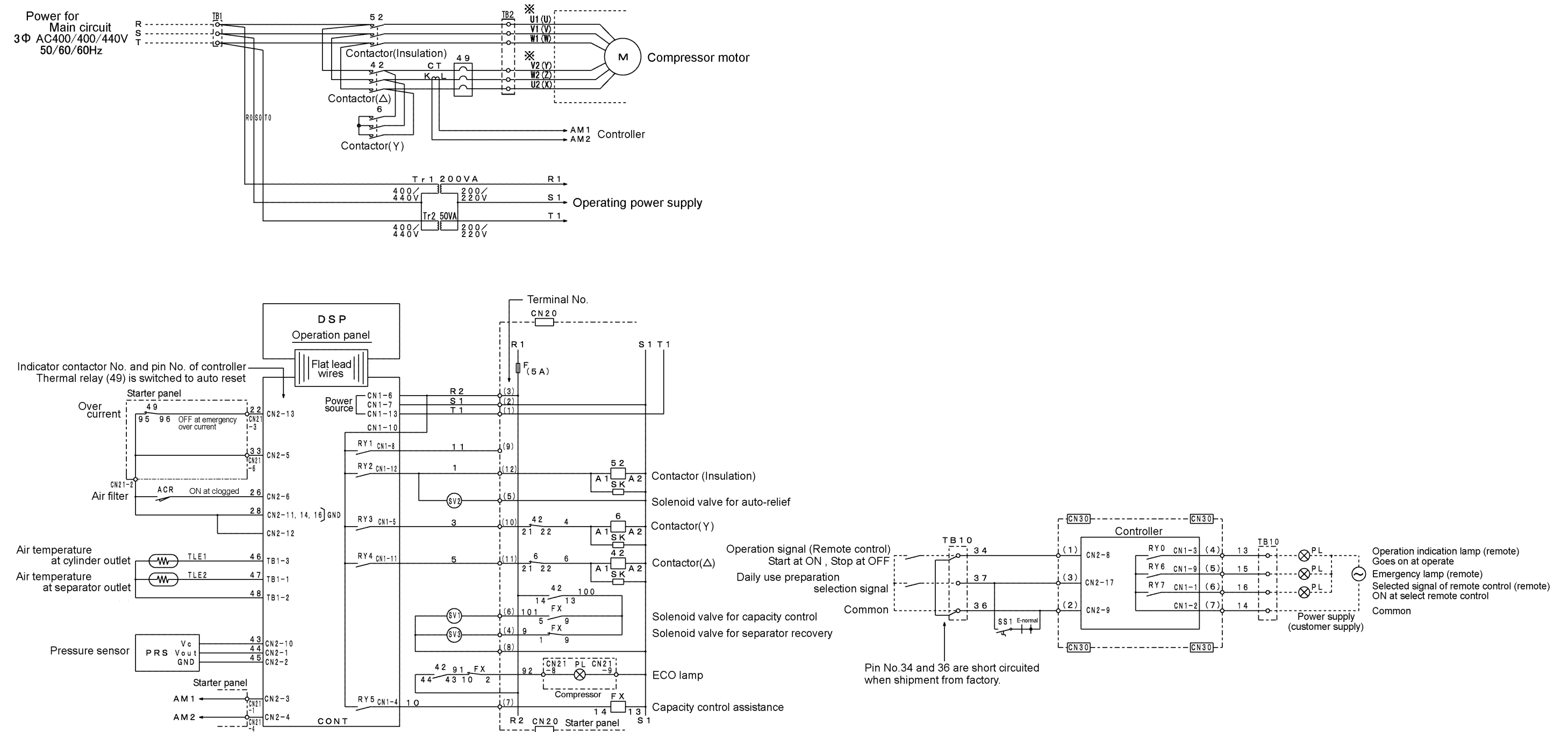
※: Such things in the parenthesis are used for the products which were produced before September, 2007.



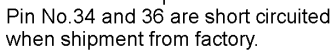
※: Such things in the parenthesis are used for the products which were produced before September, 2007.

5. Electrical System

(3) SAS22U-58/68 (400/400/440V Type)



※: Such things in the parenthesis are used for the products which were produced before September, 2007.

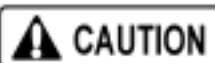


※: Such things in the parenthesis are used for the products which were produced before September, 2007.

5. Electrical System

5.4 Adjustment

5.4.1 Adjustment of discharge pressure

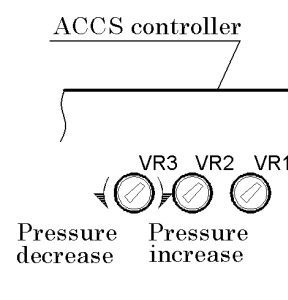


- As the discharge pressure is adjusted before shipment, do not readjust it during the normal operation.
- Do not set the unload start pressure higher than the specified setting value. Operation at higher pressure than specified set value would lead to motor overloading and motor burning and parts.

The unload start pressure (PH) and full load recovery pressure (PL) can be adjusted by the following procedures:

<Procedures>

- ① While watching the discharge pressure gauge reading, turn VR3 to adjust the full load recovery pressure. Clockwise turning increases the recovery pressure.
During adjustment, the VR2 should be kept minimized.
- ② In the same manner, turn VR2 to adjust the unload start pressure.
Clockwise turning increases the start pressure.



PK0434E

Standard setting value of pressure

Discharge pressure	0.69MPa	0.83MPa
Full load recovery pressure (PL)	0.59 MPa (6.0 kgf/cm ²)	0.74 MPa (7.5 kgf/cm ²)
Unload start pressure (PH)	0.69 MPa (7.0 kgf/cm ²)	0.83 MPa (8.5 kgf/cm ²)

IMPORTANT

- When VR3 is turned to adjust the full load recovery pressure (PL), the unload start pressure (PH) automatically changes by variations of PL. In such a case, readjust PH.
- Even when VR2 is turned to adjust the unload start pressure (PH), the PL will remain unchanged. Therefore, no readjustment of PL is required.

5.4.2 Adjustment of judged stop time

- Compressor stop time is calculated according to the pressure drop speed during auto start/stop operation and this time is a standard one to be adjusted to stop compressor.
- Turn VR1 to adjust the time. Adjustable ranges are between 30 and 150 seconds.
- When the compressor repeats start and stop frequently, adjust it to prolong the time.

Item	Standard setting	Remarks
Stop criterion time (Ts)	30 seconds	Turn VR1 fully counterclockwise.

6. Troubleshooting

6.1 How to Proceed Diagnosis

When performing failure diagnosis, pay special attention to the followings, observing general cautions.

6.1.1 Safety caution

- (1) Removing such cap and/or plug for receiver tank, fuel tanks and pipes where pressure is loaded, stop the machine and relieve all the interior pressure. Install measuring instruments connected firmly.
- (2) When doing the job with co-worker(s) together, make sure to give signal to the other person(s) and do not allow other persons to come near to the job site.
- (3) Take care not to touch hot portions and not to be involved in turning portions.

6.1.2 Caution during failure diagnosis

- (1) Do not make haste to disassemble the unit

If the unit is disassembled urgently.

- ① You may disassemble the other portions which are not related with the trouble.
- ② The cause of trouble may be missing.

The unnecessary reparations require more spare parts and man-hours, and reparation costs will increase more. What is worse, you will lose reliance or trust from clients, operators and users.

Therefore, it is absolutely necessary to investigate the trouble more carefully in advance and to follow the required procedures for failure diagnosis.

- (2) Ask the clients about the trouble in details

In order to prevent misunderstanding and incorrect judgment about the trouble, it is necessary to ask users or operators about the following questions.

- ① Is there any other disorder than the trouble he has informed?
- ② Anything abnormal occurred before this trouble?
- ③ Did this trouble happen unexpectedly? Or the unit had been operated in bad conditions before?
- ④ When and how did this trouble occur?
- ⑤ Had he repaired the unit before this trouble occurred?
- ⑥ Did he not experience similar trouble before?

- (3) Inspection items before starting diagnosis

Sometimes such trouble may be caused owing to routine mishandling of the unit. Before starting failure diagnosis, check the following items.

- ① Check whether compressor oil is lacking or dirty.
- ② Check each cable connection for any disconnection.
- ③ Check the other portions for any damage.

6. Troubleshooting

(4) Confirmation of trouble

Discuss with user(s) and/or operator(s) sufficiently about the trouble. As a result, judge whether their judgment is based on the numerical comparison or sentimental basis. Make him (them) understand well the reparation or correction you have finished.

Then check and confirm by yourself the cause of the trouble.

Note) Never proceed any investigation or measurement which may cause further greater damage.

(5) Procedures of diagnosis

When you become well experienced, you can find out the cause easily during the process of confirmation (4). But easy understanding could cause unexpected failure. So check and judge it according to the following procedures.

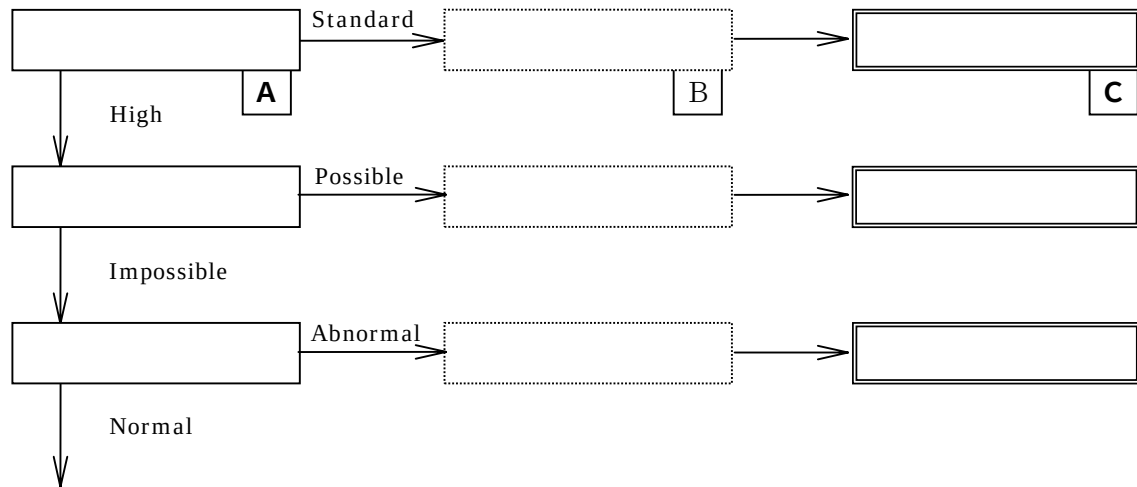
- ① Check the easiest thing or portion first.
- ② Investigate the most possible cause.
- ③ Check the other things connected to the trouble.
- ④ Check for the possibility of any other troubles.
- ⑤ Start proper and careful investigation on this trouble.

(6) Prevention of repeated occurrence of similar trouble

Even if you have repaired the trouble, unless you get rid of the fundamental cause of the trouble, it will repeatedly occur. Therefore, perform full investigation of the trouble, and it is absolutely necessary to remove the basis of the trouble.

6. Troubleshooting

6.1.3 How to use failure diagnosis table

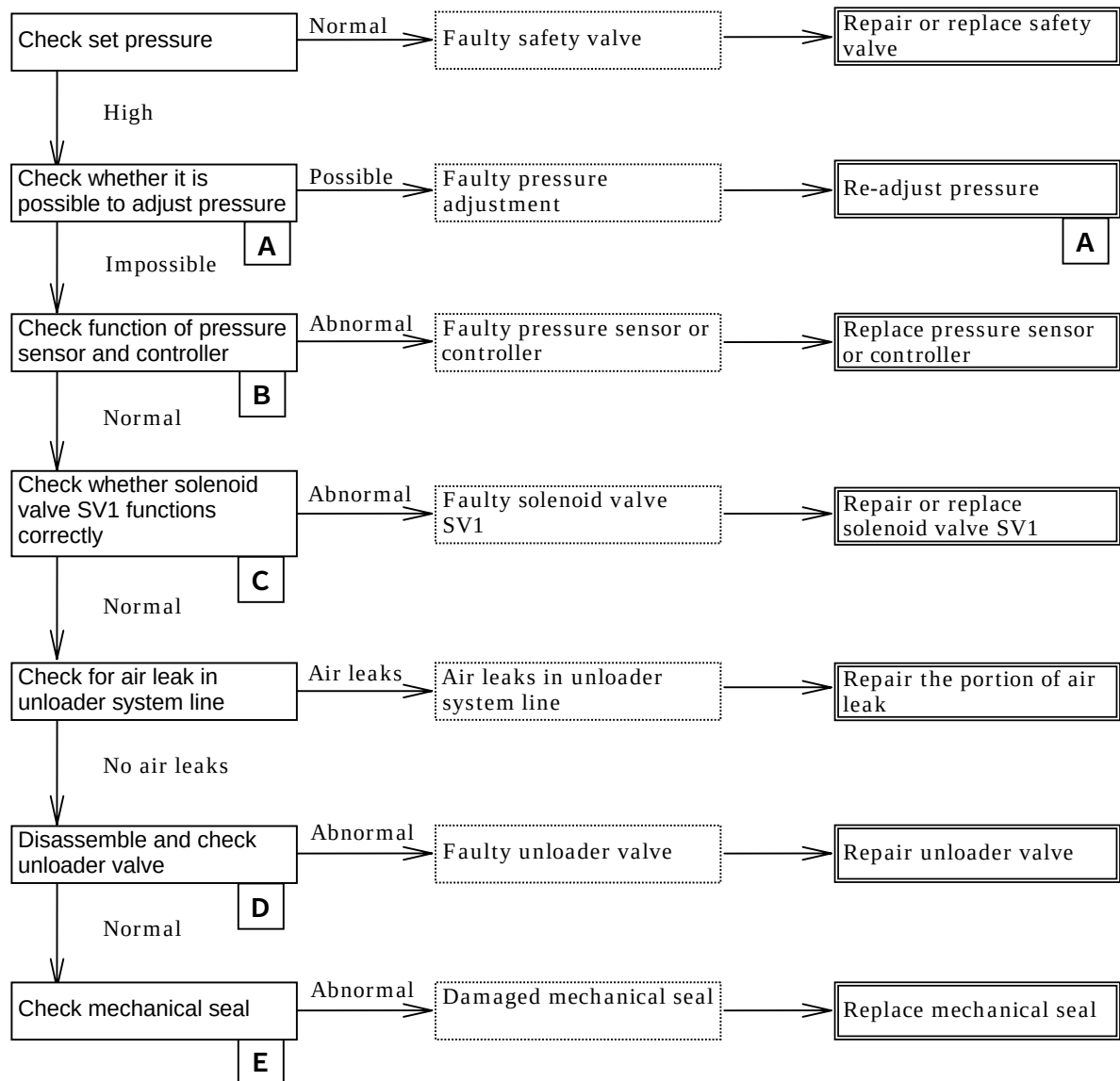


- ① In troubleshooting column The item in the column.
This item is mentioned confirmation of abnormal phenomena.
- ② In troubleshooting column The item in the column.
The cause is mentioned shows the cause of the said trouble.
- ③ In troubleshooting column The item in the column.
The measures and/or treatment shows the measures and or treatment the said cause.
- ④ **A** under each column shows the explanation of the column.

6. Troubleshooting

6.2 Failure of Capacity Control System

6.2.1 Unloader will not function and safety valve blows off



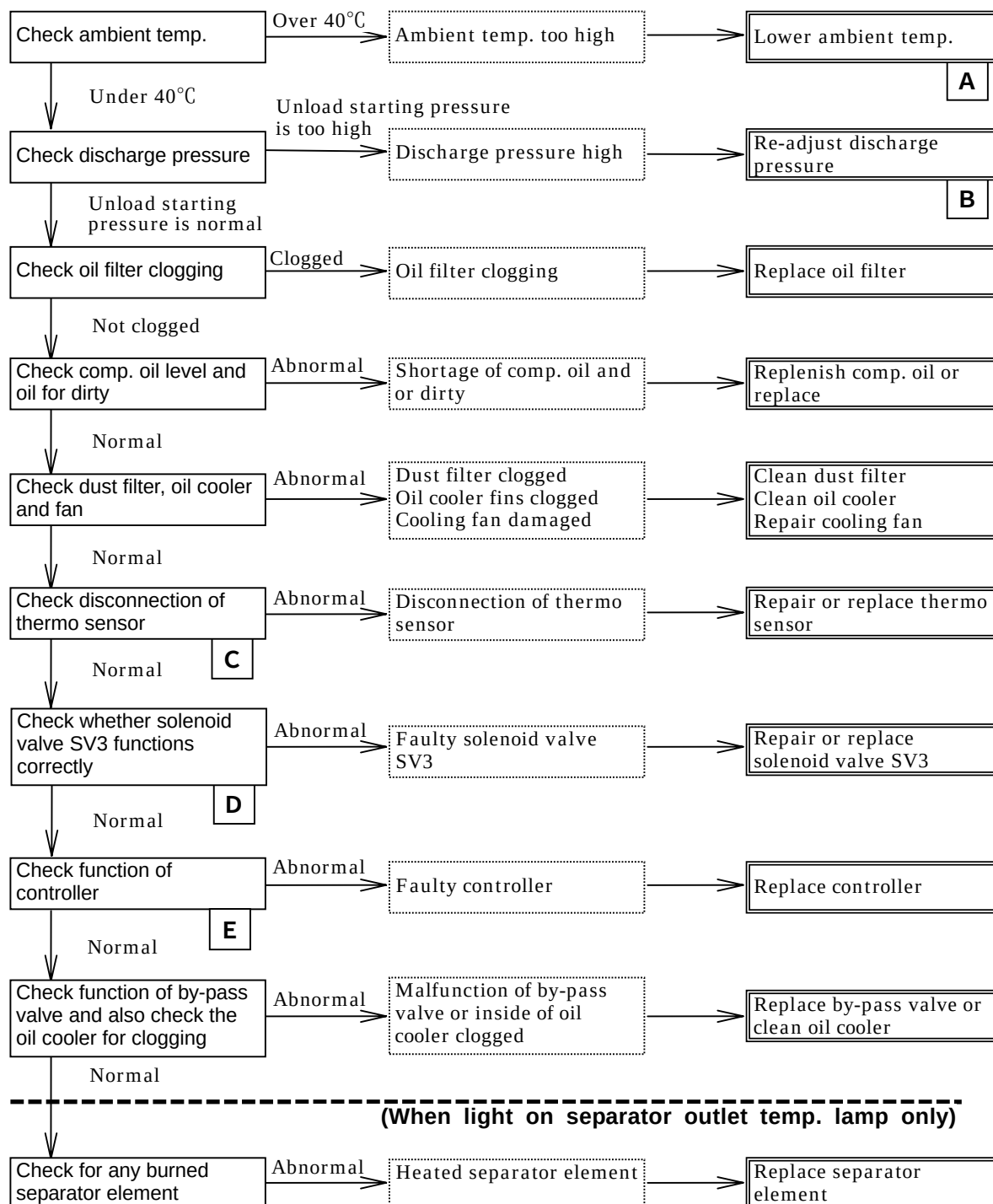
6. Troubleshooting

Mark	Item	Cause & Explanation	Countermeasures & Remarks
A	Check whether it is possible to adjust pressure	Method of adjustment...refer to 5.4.1	
B	Check function of controller	Check if the pressure sensor functions properly. (refer to 5.2.6) If it does not indicate proper signal, the pressure sensor is faulty. If the pressure sensor is correct, check if the solenoid valve SV1 shows correct volume control signal. When it starts...no conductivity is found About 10 seconds after it starts...conductivity is found After it rises to the specified pressure...no conductivity is found When no conductivity can be found even after the tank pressure exceeds 0.59MPa(6kgf/cm²) at full load or when the unloader set pressure is lowered utmost, it means owing to the defect of controller.	Check pressure sensor function Replace controller
C	Check whether solenoid valve functions correctly	Lower the unloader starting pressure to minimum during full load operation. Under these conditions, remove the pipe connected to unloader chamber, and then make sure that discharge air is delivered (when the solenoid valve is switched to no conductivity) after the pressure exceeds the pressure of about 0.59MPa (6kgf/cm²). If not, the solenoid valve is faulty.	Repair or replace solenoid valve
D	Disassemble and check unloader valve	After disassemble unloader valve, check to function of piston moving condition and valve seal condition.	If you find some abnormal function or scratch, scar, should replace damage part
E	Check mechanical seal	If air is taken in through sealing portion, unloader valve is closed, but raise tank pressure and safety valve sometimes functions.	Replace mechanical seal

6. Troubleshooting

6.3 Function of Emergency Switch

6.3.1 Cylinder outlet air temperature lamp and/or separator outlet temperature lamp glows and the machine stops



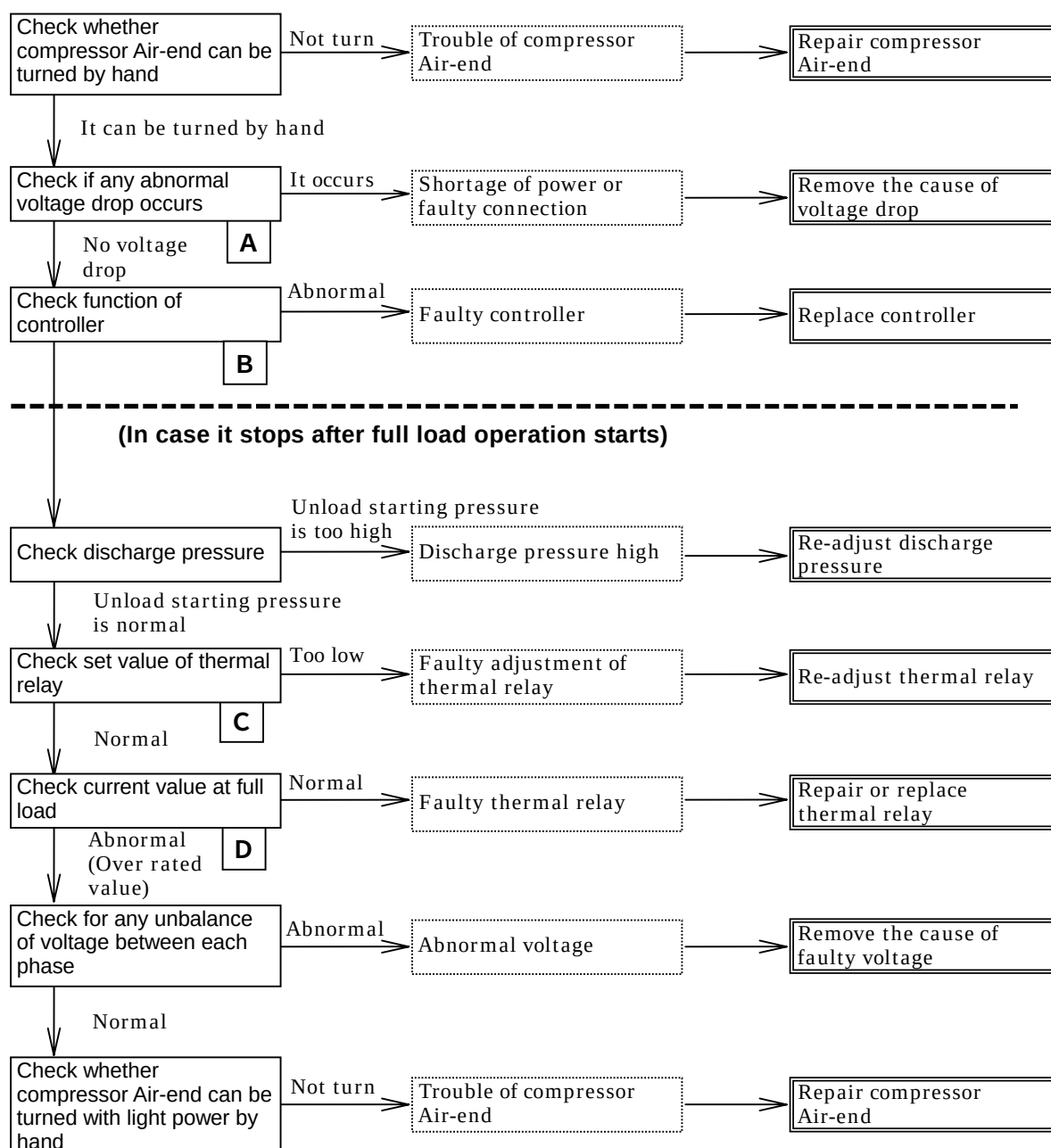
6. Troubleshooting

Mark	Item	Cause & Explanation	Countermeasures & Remarks
A	Lower ambient temp.	It sometime happen that temperature of the room is heated over 40°C by compressor installed in poorly ventilated. It causes overheating	Ventilate suction and discharge air fully of the room
B	Re-adjust discharge pressure	For adjustment...refer to 5.4.1	
C	Check disconnection of thermo sensor	It sometime happen that snapping of thermo sensor by vibration or faulty of fittings. So, check thermo sensor resistance by tester. (For the relation between temp. and resistance, refer to 4.2.3)	Repair or replace sensor
D	Check whether solenoid valve SV3 functions correctly	Make sure that the solenoid valve is switched to supply electricity and it stops separator recovery. If recovery function will not stop, the solenoid valve SV3 or electric circuit is faulty.	Repair or replace solenoid valve SV3 or electrical circuit
E	Check function of controller	Operate the compressor and check whether the discharge air temp. exceed 110°C. In case the unit stops even under normal temp., the controller is faulty. And when temp. exceeds 110°C, it is normal. Instead of resistance value when it shows disorder. (For the relation between temp. and resistance, refer to 4.2.3)	

6. Troubleshooting

6.3.2 Over current lamp glows and the unit stops

(In case it stops before full load operation starts)

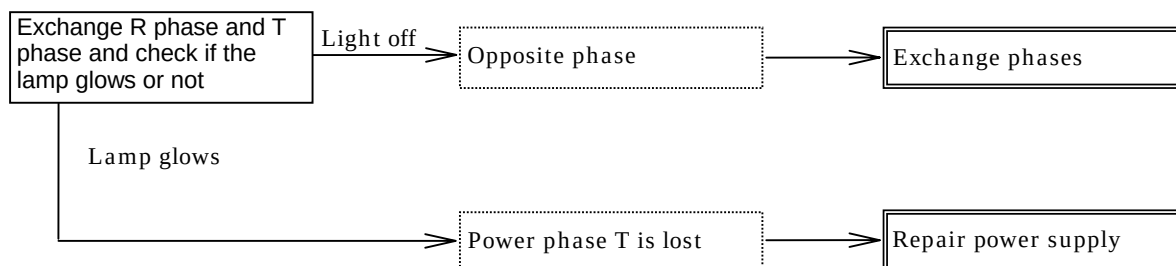


6. Troubleshooting

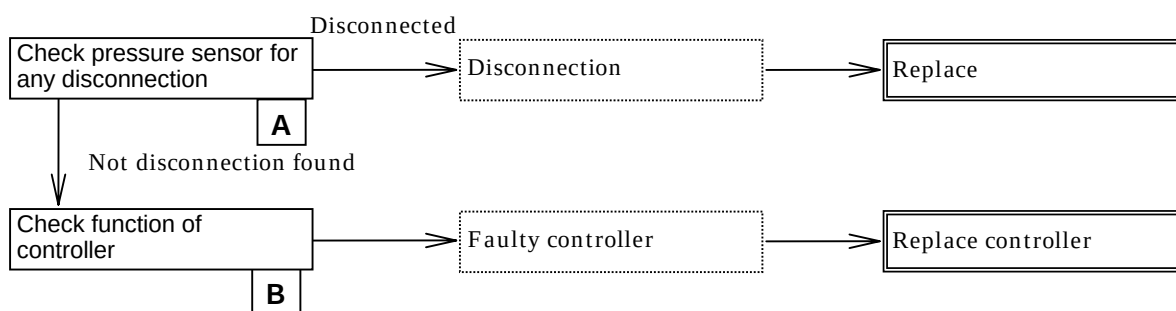
Mark	Item	Cause & Explanation	Countermeasures & Remarks
A	Check if any abnormal voltage drop occurs	Check such items as shortage of power supply, length of wiring, and cable size. They cause abnormal drop of voltage. 《Reference》 Allowable voltage drop rate shall be within 15%.	
B	Check function of controller	Check voltage between wire No.22 and 28. In case of emergency stop despite DC 0V, controller is faulty. Din case of emergency stop due to occurrence of DC12, controller is acceptable	Replace controller
C	Check set value of thermal relay	For the set value, refer to 7.1	
D	Check current value at full load	For the current value, refer to 7.1	

6. Troubleshooting

6.3.3 Opposite phase lamp glows



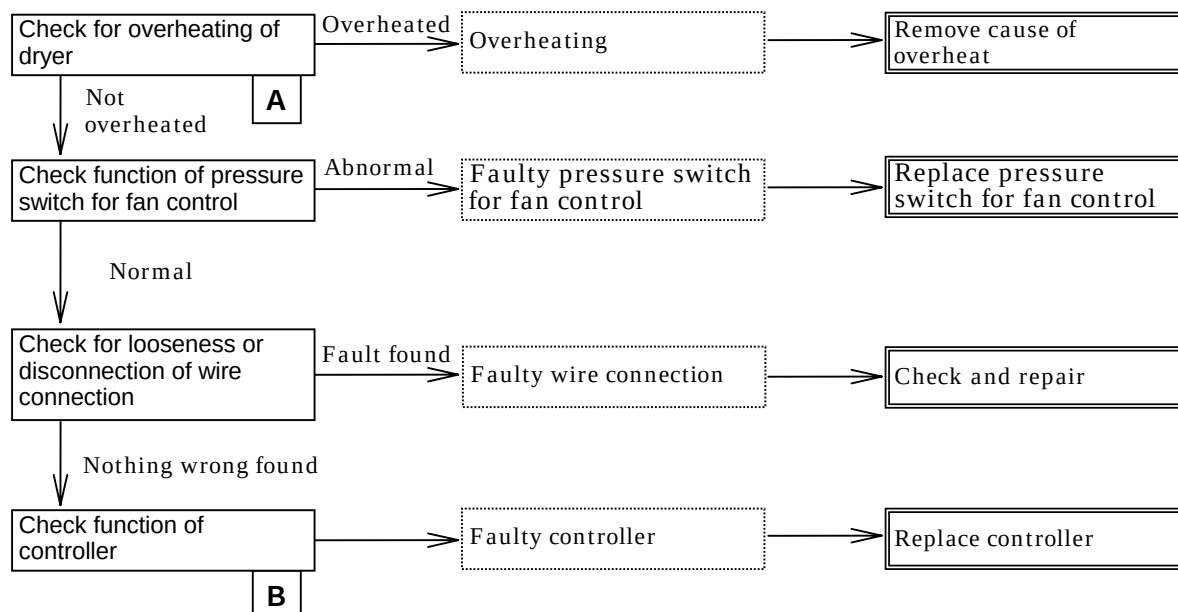
6.3.4 Pressure sensor lamp glows



Mark	Item	Cause & Explanation	Countermeasures & Remarks
A	Check pressure sensor for any disconnection	Check whether output voltage exists properly at the inlet of controller. And also check the coupler for any defects in connection.	For the output voltage, refer to 5.2.6
B	Check function of controller	If no any other causes can be guessed at all, check the controller carefully. Remove the coupler of pressure sensor, and short-circuit No.1 and 2 pin of controller CN2. Push the reset button and if it goes of, the sensor is out of order. If it will not go off, the controller is defective.	

6. Troubleshooting

6.3.5 Dryer lamp glows

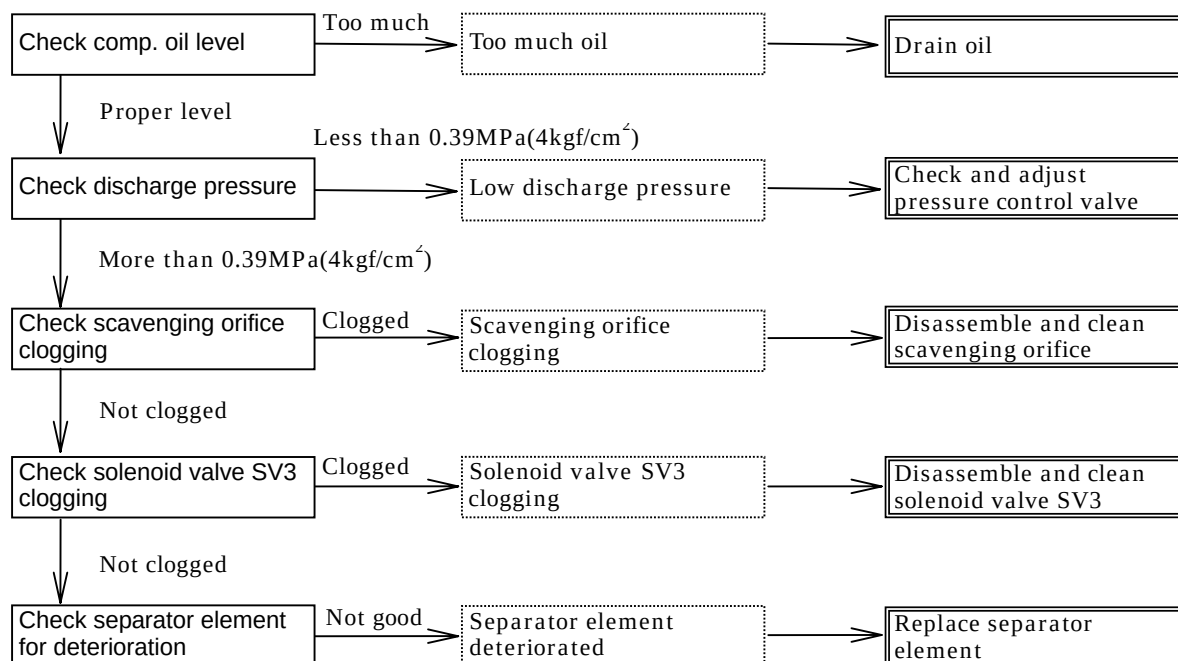


Mark	Item	Cause & Explanation	Countermeasures & Remarks
A	Check for overheating of dryer	Higher ambient temperature than 40°C and poor ventilation of dryer may cause overheating. Pay carefully attention to them especially in summer season.	Suction of air and ventilation of the room shall be sufficient To reset, release the reset button inside dryer
B	Check function of controller	In case other causes can be found, check controller. Measure voltage between cable No.33 and 28. DC0V : under normal operation DC12V: when thermal relay functions When dryer lamp cannot be reset under normal operation, controller is faulty.	Replace controller

6. Troubleshooting

6.4 Others

6.4.1 Oil is found mixed in discharge air (Oil consumption increases)



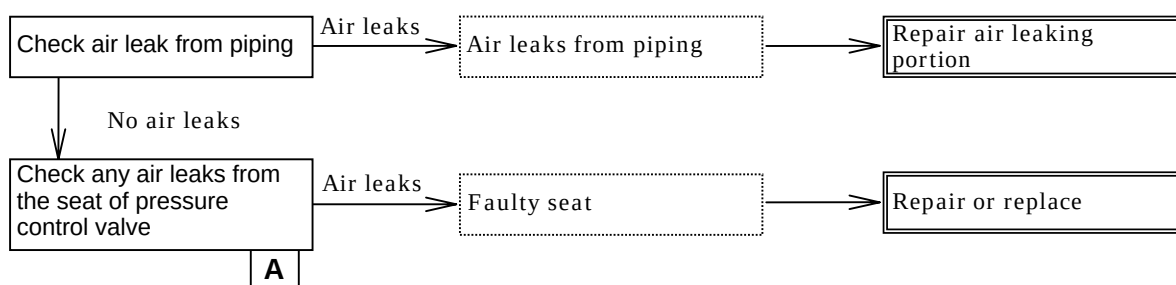
6.4.2 Abnormal noise emission from belt



Mark	Item	Cause & Explanation	Countermeasures & Remarks
A	Check belts for slipping	Abnormal noise is emitted due to slippage by primary elongation or friction	Adjust belt tension or replace

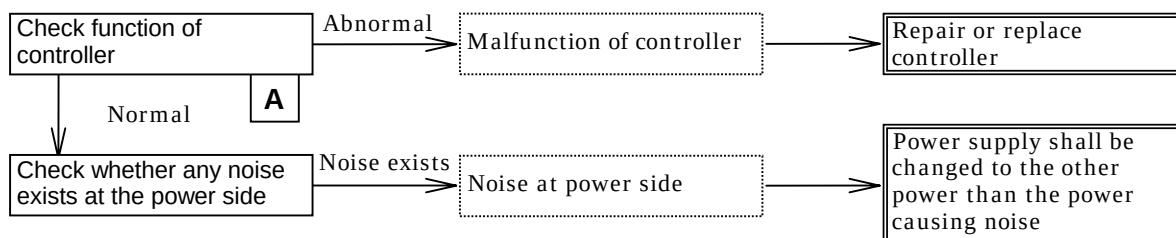
6. Troubleshooting

6.4.3 Compressor start and stops too frequently while no air is consumed (Auto-start-stop mode)



Mark	Item	Cause & Explanation	Countermeasures & Remarks
A	Check any air leaks from the seat of pressure control valve	The control valve functions as a check valve also. If air is leaking, the air in the second side is sent out through the piping of auto relief valve, and accordingly automatically starts and stops.	Replace the seat of pressure control valve

6.4.4 Compressor often stops without abnormal indicator lamp glowing and knowing no reason of the phenomena



Mark	Item	Cause & Explanation	Countermeasures & Remarks
A	Check function of controller	When performing remote control operation, check wiring of terminal No.34 and 36 for any looseness or check the remote control switch for any faulty contact.	

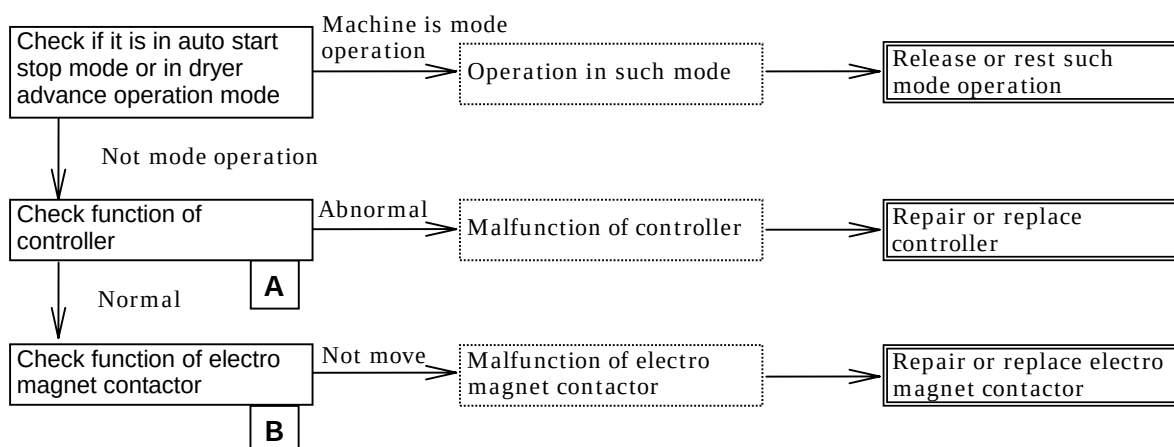
6. Troubleshooting

6.4.5 Even when each control switch is pushed ON, the machine will not start



Mark	Item	Cause & Explanation	Countermeasures & Remarks
A	Check function of controller	Check if any key is pushed due to incorrect installation of controller. If any, each operation becomes impossible. Faulty contact of each switch. Disconnection of flat cable between monitor and controller or faulty contact. Damage of controller.	Check installation of controller Repair or replace the flat cable Repair or replace controller

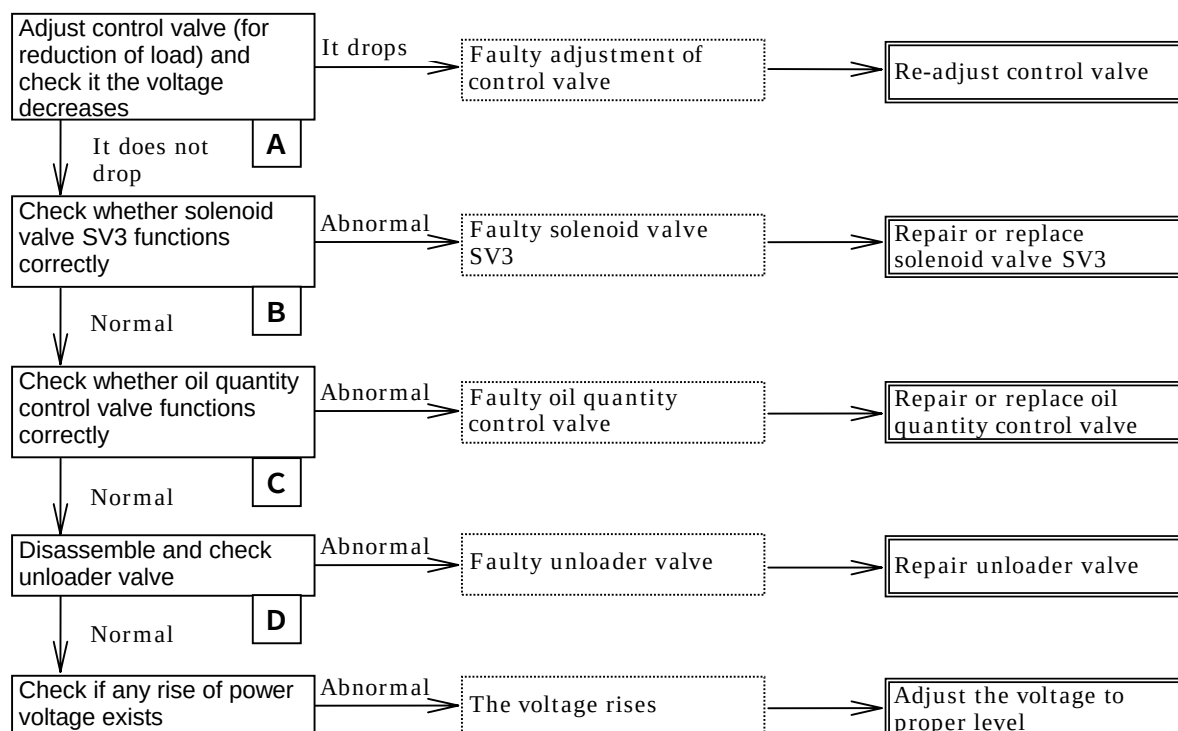
6.4.6 Start indicator lamp glows, but the unit will not start



Mark	Item	Cause & Explanation	Countermeasures & Remarks
A	Check function of controller	200V does not generate between cable No.1 ~ S1 the controller is faulty.	Replace controller
B	Check function of electromagnet contactor	If the contactor does not functions with controller working well, it is faulty or disconnection wiring. If motor does not functions with contactor moving, it is contactor or motor was disconnection or burning.	Repair or replace

6. Troubleshooting

6.4.7 Too much current flows at unloaded condition



Mark	Item	Cause & Explanation	Countermeasures & Remarks
A	Adjust control valve (for reduction of load) and check if the voltage decreases	If the control valve is kept too much open, the current value will increase. So check the relation between vacuum noise and current value. First close the valve completely and open it gradually till the vacuum noise disappears.	Readjust or replace control valve
B	Check whether solenoid valve SV3 functions correctly	Check and confirm that at unload the oil collection at the separator stops after the solenoid valve is circuited to conductivity. If the oil collection still not stop, the solenoid valve SV3 or electrical circuit is in disorder.	Repair or replace solenoid valve SV3 or electrical circuit
C	Check whether oil quantity control valve functions correctly	Check and confirm that the oil volume control valve functions properly at unload. Air comes, but it will not work. In this case, the oil volume control valve is out of order.	Disassemble it, and if any disorder is found out, it is necessary to replace parts
D	Disassemble and check unloader valve	Disassemble unloader valve and check it for malfunction of piston or any damage of sealing valve.	When any disorder is found, replace some defective part(s)

6. Troubleshooting

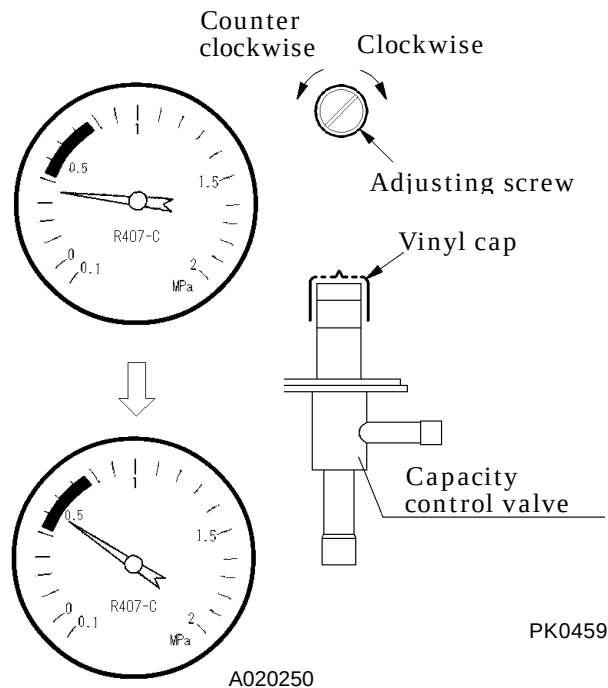
6.4.8 Troubleshooting of dryer

1. In case the interior of heat exchanger gets frozen when the dryer is operated with the indicator of refrigerant evaporating pressure gauge placed lower than the range of green zone.

- ① Stop the compressor, and melt ice.
- ② Remove the rear panel, and take off the red-vinyl cap (SAS37UD:hexagon top cap) top of capacity control valve.
- ③ Power source on and operate the dryer only, after make short circuit between the coil (wire # 11) of electric contactor (52C) for dryer and power source (wire # R2), what is installed in starter panel.

At the time, be care and be sure on safety, as the dryer is start in a moment power source ON.

- ④ Turn the capacity control valve clockwise by using (－) screwdriver so that the indicator of the refrigerant evaporating pressure gauge can be set higher than the lower limit. Soon the fan motor begins to repeat 『ON』 and 『OFF』 .
- ⑤ In case it is found impossible to adjust the capacity control valve after above adjustments have been done, the valve itself is defective. Replace it with a new one.

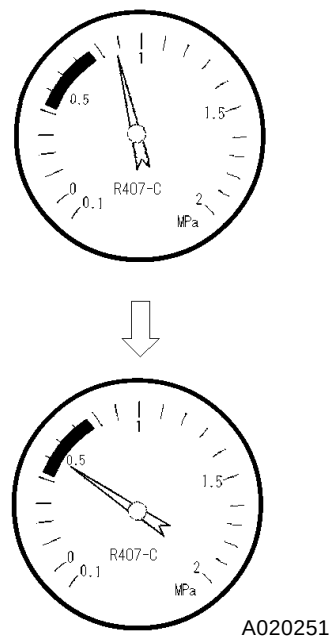


SAS11~22UD

PK0459

2. In case the dew point indicator of refrigerant evaporating pressure gauge shows unusually high temperature while the ambient temperature is not so high.

- ① Stop the compressor. After power OFF remove the panel.
- ② Remove the rear panel, and take off the red-vinyl cap (SAS37UD:hexagon top cap) top of capacity control valve.
- ③ Operate the dryer only, as same as above 1-③.
- ④ Turn the capacity control valve counterclockwise by using (－) screwdriver so that the indicator of the refrigerant evaporating pressure gauge can be set higher than the lower limit. Soon the fan motor begins to repeat 『ON』 and 『OFF』 .
- ⑤ In case proper adjustment is impossible even after above adjustments have been done, the capacity control valve itself is defective. Replace it with a new one.



SAS11~22UD

7. Reference Data

7.1 Table of Reference Data

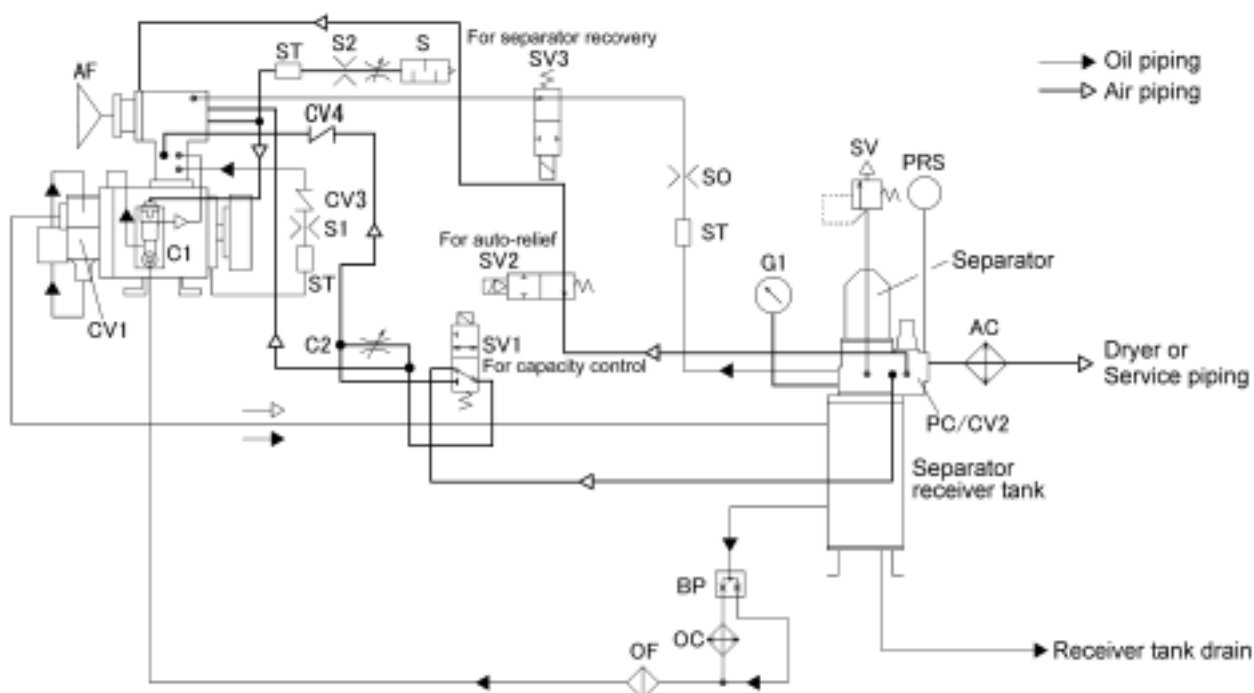
7.1.1 Table of set values

Item	Unit	SAS22UD-58/68		SAS37UD-58/68	
Voltage type		200V	400V	200V	400V
• Protection Devices					
Set value of main motor thermal relay	A	54/51/48	27/25/24	90/85/80	45/42.5/40
Set value of dryer refrigerator thermal relay	A	9/9/9		10.5/10.5/10.5	
Set value of fan for dryer condenser thermal relay	A	—		0.7/0.7/0.7	
Air temperature at cylinder outlet	℃	110			
Air temperature at separator outlet	℃	110			
• Pressure					
Air filter differential pressure switch	kPa (mmAq)	6.23 (635)			
Pressure control valve	MPa (kgf/cm ²)	0.39 (4)			
Safety valve	MPa (kgf/cm ²)	0.83 (8.5)			
Full load returning pressure	MPa (kgf/cm ²)	0.69MPa : 0.59 (6.0) 0.83MPa : 0.74 (7.5)			
Unload start pressure	MPa (kgf/cm ²)	0.69MPa : 0.69 (7.0) 0.83MPa : 0.83 (8.5)			
• Timer					
Auto start/stop. Stop judgment time	sec	30~150			
Starting unloader time	sec	5±0.2			

7. Reference Data

7.2 Piping Diagram

7.2.1 Compressor



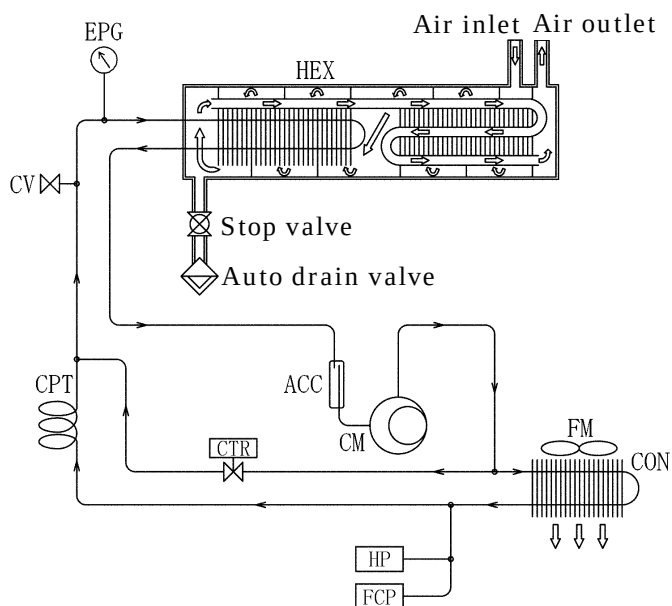
A050470E-1

AC	Aftercooler	PC	Pressure control valve
AF	Air filter	PRS	Pressure sensor
BP	By-pass valve	S	Silencer
C1	Oil control	S0,1,2	Orifice
C2	Control valve	ST	Strainer
CV1,2,3,4	Check valve	SV	Safety valve
G1	Receiver tank pressure gauge	SV1	3-way solenoid valve
OC	Pressure sensor	SV2,3	2-way solenoid valve
OF	Oil filter		

7. Reference Data

7.2.2 Dryer

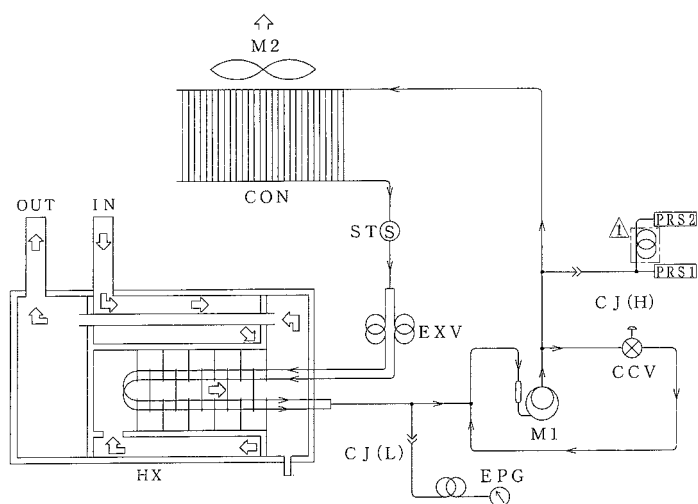
SAS22UD-58/68



A070246

HEX	Cooler
CM	Compressor
FM	Fan motor
CON	Condenser
CTR	Capacity control valve
EPG	Refrigerant evaporating pressure gauge
CPT	Capillary tube
ACC	Accumulator
HP	High pressure switch
FCP	Pressure switch for fan control
CV	Charge valve

SAS37UD-58/68



A050304

HX	Cooler
CCV	Capacity control valve
CON	Condenser
CJ(L)	Check joint at low pressure side
CJ(H)	Check joint at high pressure side
EPG	Refrigerant evaporating pressure gauge
EXV	Capillary tube
M1	compressor
M2	Fan motor
PRS1	High pressure switch
PRS2	Pressure switch for fan control
ST	Strainer

This text book contains the most recent information available at the time of printing, and the contents of the list are based on information in effect at that time and are subject to change without notice.

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